GrappStar base station build guide

10 November 2026

Subsystems: drum, frame, launch, reel, show. Model exported 2026-11-10 12:47.
All sizes in mm. Masses and print times are estimates.

1. Safety

- Wiring always starts with the battery unplugged. Plug it in last, after a check of every connection.
- Keep hands, hair and loose cloth away from the gears, the reel and the launcher while the motor can run.
- Never stand in front of the launcher. Treat it as loaded whenever the spring is cocked.
- Release the spring before you work on the launcher.
- Wear eye protection when you cut, drill or press parts, and when you test a launch.
- Let hot printed parts and the nozzle cool before you touch them. Print PPA-CF with the enclosure closed.
- Charge LiPo batteries on a fireproof surface and never leave them charging alone.

2. First print: test coupons

Print these before any big part. Each coupon uses the same geometry and numbers as the real part. Files: base/pack/coupons (3MF plates, STL, COUPON_TEST.pdf). Also print the fit coupon (python -m tools.coupon OUT) to tune cad/lib/fits.py.

Print these before the big parts. Each coupon is made from the same geometry functions and numbers as the real part, so a pass here is a pass there. Let parts cool 10 minutes before testing. The fit coupon (tolerance tiles A to H, base/pack/fit_coupon/COUPON.md) is printed alongside.

Settings

PLA: 0.4 mm nozzle, 0.2 mm layers, 4 walls, 40 % gyroid infill unless noted, 5 top and 5 bottom layers, hole and XY compensation 0, no supports. PPA-CF: hardened steel nozzle (never brass), filament dried (80 C, 8 h) and fed from a dry box, 0.2 mm layers, 6 walls, 60 % gyroid infill, 6 top and 6 bottom layers, hole and XY compensation 0, no supports. The 3MF files carry geometry only: set the material and these settings in Bambu Studio.

If a coupon fails

To change a number: on the Mac, after bash kedge/setup_mac.sh (from the repo root), cd base and run ../kedge/.venv/bin/python -m tools.risk_coupon (it rewrites base/pack/coupons and base/pack/fit_coupon). Easier: give the failing result to a Claude session and let it change the numbers. Change one number at a time and reprint only that coupon.

Plates

plate material objects
risk_plate_1_PLA.3mf PLA 24
risk_plate_2_PPA-CF.3mf PPA-CF 26

Tests

1. Swing-arm direction splitter

- Parts: rc1_plate, rc1_sleeve, rc1_sun, rc1_arm, rc1_drag_pad, rc1_drag_shim, rc1_idler_clip, rc1_idler, rc1_output, rc1_crank, rc1_stop, rc1_stop_e
- Bought: 4 x M3x16 (two stop fingers); everything else printed
- Do: Push the drag pad root block and the shim into the arm hub pocket (finger in its recess). On the plate: sleeve on the centre pin, sun on its rib, arm on the sleeve above the sun with its pin down through the idler, printed C clip into the pin groove, output on its pin, crank on top, the two stop fingers on their columns (2 x M3x16 each). Turn the crank 10 turns each way.
- Pass: Clockwise seen from above: the arm swings the idler into the output at once and comes to rest on the engaged stop finger, the output turns, no skipped teeth. Brake the output lightly with a finger and keep turning: the arm stays on that finger (the idler is held at full depth, never pulled in deeper) and the crank still turns smoothly, no binding. Anticlockwise: the arm swings out to the other finger within about 20 deg of crank, the idler runs free and the output stays still. No jam at engagement. The pad drags the arm reliably both ways and the clip stays on.
- If it fails, change: launch_aim.py: FIT (pin and bore play), DRAG_INT and DRAG_SHIM_ADJ (drag: or print a thicker shim), clips.py for the C clip, A() alpha_d = alpha_e + 20 (swing out), _stop_post_xy (swung-out stop), _stop_post_e_xy (engaged stop: binding with the output braked means it lets the idler in too deep); launch_gears.py center_distance extra / spur backlash (mesh).

- Made from: launch_aim.sun_solid, arm_solid, idler_solid (m1.5 z30/18/30, face 12, FIT 0.35), launch_gears.spur for the output, launch_aim.A: centre distances 36.10 / 36.10, output at (11.7, 68.4), engaged arm 96.2 deg on the engaged stop (launch_aim._stop_post_e_xy), swung out 116.2 deg on the other stop (launch_aim._stop_post_xy), sleeve rib launch.RIB_W 5.8 / RIB_R 12.6; drag_pad_solid, drag_shim_solid (DRAG_INT 0.45, shim up to +0.25), pin_clip_solid

2. Printed contact switch

- Parts: rc2_switch_base, rc2_switch_lever

- Bought: M3x10 SHCS ISO 4762 + M3 jam nut, M3 nut, 2x M3x10 self tapping (pan or SHCS) through the lever root into the pad, at most M3x11.7, 2 wires, multimeter

- Do: Nut into the pillar trap (drop of glue), lever clamped with 2 M3 self tappers, contact screw head down through the lever, set the head 0.9 mm above the nut (0.5 mm per turn), lock the jam nut. Meter on beep between screw and nut. Press the lever tip 100 times.

- Pass: Measure the tip travel with calipers (depth rod from the lever tip down to the base top, note the reading at rest, press until it beeps, read again): beep within 0.8 to 1.4 mm of tip travel (design 1.14), silent when let go, tip stops on the hard stop without cracking, lever back flat, same after 100 presses.

- If it fails, change: cad/lib/switch.py contact_switch args travel, overtravel, t, b; STRAIN_LIMIT, NUT_PRESS (loose nut), SCREW_LENGTHS.

- Made from: cad.lib.switch.contact_switch(lever_len=36) with the library defaults (travel 0.8, overtravel 0.6, beam t 1.6): gap 0.91, pretravel at the tip 1.14, close force 0.83 N, strain 0.0061

3. Dog clutch and compliant fork

- Parts: rc3_base, rc3_reel_face, rc3_collar, rc3_fork, rc3_dog_post, rc3_rail

- Bought: M3 nut (yoke tab), 2 x M3x12 (SW_DOG screws in the post), 1 x M3 (post clamp), multimeter

- Do: Press the rail into the base. REEL face on the journal (dogs up), collar on the hex (dogs down), fork pads in the collar groove and its sleeve on the rail, dog post clamped on the rail below the fork, SW_DOG screws in the post set 0.5 mm short of the tab nut at full engagement. Push the fork 7 mm down with the dogs tip to tip, then turn the REEL face until they drop in. Turn the face hard both ways, then pull the fork back. Meter on beep between the lower screw and the tab nut.

- Pass: Yoke flexes when tip to tip (no jam, roofs slide off), dogs drop in within a quarter turn, under hand torque the dogs pull in, never walk out, a light pull on the fork releases them, the yoke springs back flat; the meter beeps only with the dogs fully in, never tip to tip.

- If it fails, change: _drum_kin.py DEFAULTS dog_backcut, dog_roof, dog_deg, dog_gap, fork_k; drum.py GEOM fork_t, yoke_r, groove, dog_touch, dogpost_s; fits.py SHAFT_IN_BORE (collar on the hex, face on the journal), HOLE_SLIDE (fork on the rail).

- Made from: drum.dog_teeth (dog_n 4, dog_h 5, dog_deg 36, backcut 4, roof 1.2, r 18..26), drum.fork, drum.dog_post (dog_touch 0.5), drum.rail, drum.hexprism, GEOM journal d33, groove (82.0, 88.0), stroke 7, neutral gap 2.5

4. Face slip clutch with the flexure spring

- Parts: rc4_hub_plate, rc4_friction_disc, rc4_pressure_plate, rc4_flexure, rc4_adjuster_plate

- Bought: 3 x M3x20 SHCS, 3 x M3 nut, string, coins in a small bag, kitchen scale

- Do: Stack on the key stub: hub plate (counterbores down), friction disc, pressure plate (pads up), flexure (beam down, tabs on the screws), adjuster plate (traps up); 3 screws from below, nuts on top. Run the nuts down until the flexure just touches (zero), then turn all 3 nuts the same, 1/4 turn (0.125 mm) at a time. Clamp the stub level in a vice, tie a string to a disc lug (r 28.5 mm), lay it over the rim and hang a bag of coins weighed on a kitchen scale. Add or take coins to find the hold and slip weights at each setting.

- Pass: At each setting the lower weight must not pull the disc round and the higher one must (smooth steady slip, no judder); same result 3 times; the flexure never bottoms before 1 turn. Bands: 1/4 turn: 35 g holds, 60 g slips (design 48 g, 0.013 N m); 1/2 turn: 70 g holds, 120 g slips (design 95 g, 0.027 N m); 3/4 turn: 105 g holds, 180 g slips (design 143 g, 0.040 N m); 1 turn: 140 g holds, 240 g slips (design 190 g, 0.053 N m).

- If it fails, change: cad/lib/clutch.py: slip_clutch args flex_t, flex_gap, flex_ro, r_in, r_out, n_grooves; MU_PLA_PLA, E_PLA, SIGMA_FLEX_PLA_SUSTAINED (creep: if the torque falls overnight).

- Made from: cad.lib.clutch.slip_clutch() defaults (reel does not use it: its clutch is the spool face plus the PETG rocker): shaft d20 key 6, friction r 18.5..27, mu 0.3, flexure t 0.8 gap 0.5, 0.106 N m per mm, travel 0.50; on the shared keyed stub of coupon 7

5. Bearing seats

- Parts: rc5_seats_608_693, rc5_csk20_seat

- Bought: 608ZZ, F693ZZ, 693ZZ, CSK20PP

- Do: Press each bearing in square with a flat block or a vice (tile left to right: 608, F693ZZ, 693ZZ); CSK20PP into the round boss with its outer keyway on the printed key.

- Pass: Firm press, seated flat on the lip, stays in when tapped upside down, spins as freely as before; the CSK20PP key slides in without force, it still freewheels one way and locks the other.

- If it fails, change: fits.py BRG_RIB_INTERF, BRG_RIB_BORE (all seats); bearings.py CSK20_KEY_RCLEAR, CSK20_OKEY_W (key), KEY_CLEAR in fits.py. If the 608 is off, print tile A of the fit coupon (base/pack/fit_coupon, COUPON.md).

- Made from: cad.lib.bearings.seat_608(bottom lip, lip_d 14) as reel, seat_f693(), seat_693(), seat_csk20(key_deg 90) as the reel capstan block (wall 4); fits BRG_RIB_INTERF 0.15, BRG_RIB_BORE 0.3, CSK20_KEY_RCLEAR 0.2

6. M3 holes

- Parts: rc6_m3_tile

- Bought: M3x4x5 heat-set insert, M3 nut, 2 x M3x10

- Do: Left to right: heat-set the insert (220 C), press the nut in, push a screw through, self-tap a screw in.

- Pass: Insert flush and does not spin when the screw is tightened; nut presses in and holds; screw drops through the clearance hole; the tap hole cuts a thread and holds a hard pull without splitting.

- If it fails, change: fits.py M3_INSERT_D, M3_NUT_AF, M3_CLEAR_D, M3_TAP_D (tiles C, D, E of the fit coupon in base/pack/fit_coupon bracket them).

- Made from: cad.lib.fasteners.m3_insert (d 4.0), m3_nut_trap (AF 5.8), m3_clear (d 3.4), m3_tap (d 2.6)

7. Key rib and spigot

- Parts: rc7_key_stub, rc7_key_bore, rc7_spigot, rc7_bushing

- Bought: CSK20PP from coupon 5 (inner keyway)

- Do: Slide the key stub into the keyed tile and into the CSK20PP inner race (the real interface). Push the spigot into the sleeve slice and turn it.

- Pass: Stub slides in by hand, key in the keyway, no visible rock. Play: clamp the tile, tape a ruler to the stub end, mark the ruler 60 mm from the axis, rock the stub both ways: the mark moves under 1 mm (about 1 deg). The spigot turns freely; rocked at its free end it moves under 0.5 mm.

- If it fails, change: fits.py KEY_CLEAR, SHAFT_IN_BORE; shafts.py KEY20_H, SHAFT_PRINT_UNDERSIZE; launch.py SPIGOT_BORE (now 12.35); drum.py GEOM spig_d.

- Made from: cad.lib.shafts.keyed_shaft(d 20, key_w 6, key_h 2.6) and keyed_bore (clear 0.3); drum.main_shaft (spig_d 12); launch.sleeve_solid (SPIGOT_BORE 12.35)

8. Gear meshes m1.25, m1.5, m2, m2.5

- Parts: rc8_m125_plate, rc8_m125_seg, rc8_m125_pinion, rc8_m15_plate, rc8_m15_z42, rc8_m15_z24, rc8_m2_plate, rc8_m2_z50, rc8_m2_z22, rc8_m25_plate, rc8_m25_big, rc8_m25_drive

- Bought: none

- Do: Each pair on its plate (label = module), flat faces down; turn or rock through every tooth.

- Pass: Smooth, no tight spot, a little play when rocked; m2.5: the tall tooth and the cut-down last tooth pass without catching, then the mesh lets go at the gap.

- If it fails, change: fits.py GEAR_BACKLASH, GEAR_CENTER_EXTRA (m1.25, m1.5, m2); launch_gears.py center_distance extra, spur backlash, sector_gear_2d tall_factor (m2.5); _drum_kin.py pin_x (m1.25 pinion).

- Made from: m1.25: gears.spur_gear z60 sector / z10 shift 0.4 tip_short 0.08 (drum.tip_short_for), cd 44.33 (drum.a_seg); m1.5: gears.spur_gear z42 sector / z24, cd 49.60; m2: gears.spur_gear z50 sector (drum GEOM out_m, out_z) / z22 (reel Z_CAPG), cd 72.10; m2.5: launch_gears.sector_gear_2d z40 (n_present 27, last 6 + gap) / launch_gears.spur z26, cd 82.60 (launch L); coupon face 6, fits GEAR_BACKLASH 0.15, GEAR_CENTER_EXTRA 0.1

9. Shear-pin torque limiter

- Parts: rc9_hub_lever, rc9_outer_lever, rc9_shear_pin x3

- Bought: string or a strap, luggage scale

- Do: Print the pins lying down on their flat, as exported (D44: shear plane across the layers; never stand them up). Hub slice into the outer slice, pin through both holes (head outside). Clamp the hub lever, hang the luggage scale on the outer lever hole (r 60 mm) at right angles and pull slowly until the pin shears. Break all 3 pins. Torque = kg x 9.81 x 0.06 N m.

- Pass: Each pin shears cleanly between 5.9 and 10.2 kg on the scale (3.5 to 6 N m, drum_NOTES; design 4.9 N m = 8.3 kg), the 3 pins within 20 percent of each other, both slices undamaged (holes still round); a fresh pin goes in by hand.

- If it fails, change: drum.py GEOM shear_d (torque goes with d squared), hub_od (radius), shear_s; fits for the hole: drum.coupler shear hole (shear_d + 0.1) and shear_pin (shear_d - 0.04). Spec D8 / D26 limit 5 N m.

- Made from: drum.coupler_hub and drum.coupler (sliced s 4.5..20), drum.shear_pin: d4 PLA in single shear at r 13 (shear_s 9); 4.9 N m at 30 MPa

10. AS5600 magnet cap (bench test)

- Parts: rc10_magnet_cap

- Bought: magnet: not included with the board, see MEASURE_LIST C13 (any small neodymium disc edge-on, or 6 x 2.5 diametric), M3x6, superglue

- Do: Glue the magnet into the end pocket (flush, any way round), push the cap on the shaft end (flat to the flat), M3x6 onto the flat. It fits the d12 main shaft spigot of the base (for the bare-motor bench test, tape the magnet to the wiper shaft end instead). AS5600 chip on the shaft axis, 1 to 2 mm from the magnet face.

- Pass: Cap does not wobble or slip on the shaft; the AS5600 angle follows the shaft smoothly through full turns (no jumps), magnet status OK (not too weak, not too strong) at 1 to 2 mm.

- If it fails, change: drum.py GEOM cap_s, spig_d; magnet_holder pocket d6.2 x 2.6 (magnet loose or tight); fits.py HOLE_SLIDE (cap on the spigot).

- Made from: drum.magnet_holder (spig_d 12 with its flat, cap_s (195.0, 201.5), magnet pocket d6.2 x 2.6), printed as drum prints it

Parts

part material qty print bbox mm what
rc1_plate PLA 1 110.1 x 82.2 x 33.1 Splitter base: centre pin d8, output pin d8, two stop finger columns (2 x M3 tap each)
rc1_sleeve PPA-CF 1 26.0 x 26.0 x 29.3 Sleeve stub d20 with the integral key rib (launch.RIB_W, RIB_R) under the sun and the crank
rc1_sun PPA-CF 1 47.8 x 48.0 x 12.0 launch_aim.sun_solid: sun m1.5 z30 as built
rc1_arm PPA-CF 1 61.4 x 41.0 x 19.3 launch_aim.arm_solid: swing arm as built (drag pad pocket, d8 pin, clip groove)
rc1_drag_pad PLA 1 24.6 x 8.7 x 4.0 launch_aim.drag_pad_solid: printed flexure drag pad as built
rc1_drag_shim PLA 1 9.4 x 3.0 x 4.0 launch_aim.drag_shim_solid: nominal preload shim as built
rc1_idler_clip PLA 1 10.8 x 10.0 x 1.5 launch_aim.pin_clip_solid: printed C clip for the d8 idler pin
rc1_idler PPA-CF 1 30.0 x 29.7 x 12.0 launch_aim.idler_solid: idler m1.5 z18 as built
rc1_output PPA-CF 1 47.8 x 48.0 x 12.0 Output m1.5 z30 (launch_gears.spur, face 12) on a d8 pin
rc1_crank PPA-CF 1 48.2 x 30.3 x 20.0 Hand crank, keyed on the sleeve rib like the sun
rc1_stop PLA 1 47.1 x 10.0 x 6.4 Stop finger: rounded end at the as-built swung-out stop post position
rc1_stop_e PLA 1 64.4 x 10.0 x 6.4 Engaged stop finger: rounded end at the as-built engaged stop post position (launch_aim._stop_post_e_xy)
rc2_switch_base PLA 1 54.0 x 35.0 x 11.5 contact_switch base: pad, nut pillar, hard stop, wire channel
rc2_switch_lever PLA 1 50.0 x 5.0 x 8.0 contact_switch lever, printed on its side (Y up)
rc3_base PLA 1 40.0 x 91.5 x 97.5 Base: d30 journal (REEL sleeve) and hex 22 AF (main shaft), rail hole (press)
rc3_reel_face PPA-CF 1 61.6 x 54.0 x 9.5 REEL gear dog face (drum.dog_teeth, phase 45) with a handle
rc3_collar PPA-CF 1 60.0 x 60.0 x 18.0 Collar lower half: drum.dog_teeth facing down, fork groove, hex bore

rc3_fork PLA 1 87.1 x 94.1 x 83.0 drum.fork as built: compliant yoke, rail sleeve, follower pin, M3 nut in the tab
rc3_dog_post PLA 1 20.0 x 75.9 x 41.7 drum.dog_post as built: SW_DOG_L / SW_DOG_R screw bosses, clamps on the rail
rc3_rail PPA-CF 1 110.0 x 10.0 x 9.0 drum.rail profile d10 with the print flat, cut to 110 long
rc4_hub_plate PLA 1 54.0 x 54.0 x 6.0 clutch.slip_clutch hub_plate (library defaults)
rc4_friction_disc PLA 1 57.0 x 55.0 x 3.0 clutch.slip_clutch friction_disc (library defaults)
rc4_pressure_plate PLA 1 54.0 x 54.0 x 3.5 clutch.slip_clutch pressure_plate (library defaults)
rc4_flexure PLA 1 50.0 x 50.0 x 1.7 clutch.slip_clutch flexure (library defaults)
rc4_adjuster_plate PLA 1 54.0 x 54.0 x 4.0 clutch.slip_clutch adjuster_plate (library defaults)
rc5_seats_608_693 PLA 1 60.0 x 28.0 x 9.6 608 (reel spool seat), F693ZZ and 693ZZ crush-rib seats
rc5_csk20_seat PLA 1 55.3 x 55.3 x 16.0 CSK20PP seat with the printed key (reel capstan block, key_deg 90)
rc6_m3_tile PLA 1 54.0 x 18.0 x 8.6 M3 heat-set hole, nut trap, clearance and tap holes (fits.py values)
rc7_key_stub PPA-CF 1 21.2 x 20.0 x 31.0 shafts.keyed_shaft d20 key 6 (reel capstan tube numbers), 31 long, also the coupon 4 clutch shaft
rc7_key_bore PPA-CF 1 36.0 x 36.0 x 8.6 shafts.keyed_bore d20 key 6, 8 long (keyed hubs on the sleeve rib are PPA-CF)
rc7_spigot PPA-CF 1 28.0 x 25.4 x 22.0 drum.main_shaft slice: hex 8 long + d12 spigot 20 long, on a hex flat
rc7_bushing PPA-CF 1 32.0 x 32.0 x 16.0 launch.sleeve_solid slice 16 long: spigot bore d12.35
rc8_m125_plate PPA-CF 1 62.3 x 18.0 x 11.0 Pins at 44.33
rc8_m125_seg PLA 1 38.1 x 46.7 x 6.0 drum segment gear m1.25 z60 sector
rc8_m125_pinion PPA-CF 1 15.8 x 15.2 x 6.0 drum wrap pinion m1.25 z10 x0.4 tip_short 0.08
rc8_m15_plate PPA-CF 1 67.6 x 18.0 x 11.0 Pins at 49.60
rc8_m15_z42 PPA-CF 1 41.0 x 29.6 x 6.0 drum wrap idler output m1.5 z42 sector
rc8_m15_z24 PPA-CF 1 38.9 x 38.9 x 6.0 reel cup ring gear m1.5 z24
rc8_m2_plate PPA-CF 1 90.1 x 18.0 x 11.0 Pins at 72.10
rc8_m2_z50 PPA-CF 1 61.0 x 39.3 x 6.0 drum REEL output m2 z50 sector
rc8_m2_z22 PPA-CF 1 48.0 x 47.7 x 6.0 reel capstan gear m2 z22
rc8_m25_plate PPA-CF 1 100.6 x 18.0 x 11.0 Pins at 82.60
rc8_m25_big PPA-CF 1 61.5 x 49.0 x 6.0 launch big gear m2.5 z40 sector: teeth 21..26 (tall, relieved)
rc8_m25_drive PPA-CF 1 42.5 x 44.0 x 6.0 launch drive gear m2.5 z26 sector
rc9_hub_lever PPA-CF 1 79.0 x 26.0 x 15.5 drum.coupler_hub slice (shear pin hole) with a lever, string hole at r 60
rc9_outer_lever PPA-CF 1 86.0 x 40.0 x 9.5 drum.coupler slice (shear pin hole) with a lever, string hole at r 60
rc9_shear_pin PLA 3 12.1 x 5.0 x 4.1 drum.shear_pin as built (d4 PLA), printed LYING on its 0.4 mm flat as drum prints it (D44): the shear plane runs across the layers
rc10_magnet_cap PLA 1 20.0 x 20.0 x 6.5 drum.magnet_holder as built: AS5600 magnet cap on the d12 spigot end

3. Parts list

Printed parts

part	subsystem	material	qty	mass g	notes
drum_bracket_a	drum	PLA	1	37.7	Drum bracket A: axle, shift rail, N20 ear, 2 x M3 to the deck
drum_bracket_b	drum	PLA	1	23.4	Drum bracket B: axle, shift rail, 2 x M3 to the deck
drum_bracket_idler	drum	PLA	1	12.5	Wrap idler journal post, 2 x M3 to the deck
drum_bracket_idler2	drum	PLA	1	16.2	Wrap idler second journal post (motor end), 2 x M3 to the deck
drum_drum_a	drum	PLA	1	146.5	Drum A: N20 gear 120T m0.8, brake cam, wrap segment gear + locking arc, hex spigot
drum_drum_b	drum	PLA	1	79.0	Drum B: shift groove barrel, Gray code bands A B C, hand lever hex boss
drum_fork	drum	PLA	1	25.9	Compliant shift fork: thin yoke is the spring (about 1.5 N/mm), follower pin in groove 1, contact tab
drum_hand_lever	drum	PLA	1	4.0	Hand lever on the drum hex boss (fallback: swing the N20 out first)
drum_n20_cradle	drum	PLA	1	11.9	N20 cradle on the bracket A ear: pivot + swing screw (hand fallback), SW_N20 contact
drum_n20_strap	drum	PLA	1	2.7	End plate closing the N20 cradle, 2 x M3 x 8
frame_can_stay	frame	PLA	1	28.4	can stay: post on the motor plate (2 x M3 x 12 into plate inserts), cable tie or hose clamp round the can through the cradle slots; reprint with the adapter

part	subsystem	material	qty	mass g	notes
frame_cheek_r	frame	PLA	1	197.7	motor plate (y -10..0, front face = the spec's y 0): window for the puck adapter, 8 rim holes (M3 x 12 from the front into the puck lip), bridge to the motor leg (2 x M3 x 12), can stay and hood inserts; carved round the other subsystems
frame_deck_1	frame	PLA	1	182.1	deck: M3 inserts for every frame foot; 9 x M3 rover holes (it is its own drill template)
frame_deck_2	frame	PLA	1	349.5	deck: M3 inserts for every frame foot; 14 x M3 rover holes (it is its own drill template)
frame_deck_3	frame	PLA	1	189.9	deck: M3 inserts for every frame foot; 9 x M3 rover holes (it is its own drill template)
frame_deck_4	frame	PLA	1	362.4	deck: M3 inserts for every frame foot; 14 x M3 rover holes (it is its own drill template)
frame_deck_ext	frame	PLA	1	340.5	deck: M3 inserts for every frame foot; 8 x M3 rover holes (it is its own drill template)
frame_guard_front	frame	PLA	1	46.1	front gear guard between the drum gears and the bay, M3 x 10 into the deck
frame_guard_hood	frame	PLA	1	26.2	gear hood over the reel and bus gears, M3 x 10 through its lugs into the motor plate
frame_motor_leg	frame	PLA	1	117.3	motor support leg (prints upright): 4 x M3 x 10 into the deck, 2 x M3 x 12 along y into the bridge behind the motor plate
launch_aim_cap	launch	PLA	1	5.2	Worm shaft saddle cap, 4 x M3x30 through the legs into 13 deep pilots in the saddles
launch_aim_crankpin_clip	launch	PLA	1	0.1	Printed C clip on the crank pin, inboard of the conrod
launch_aim_housing	launch	PLA	1	185.4	Aim housing on the deck (4 x M3 x 10 from the top through its counterbored foot flanges into the deck inserts): sleeve end bushing, countershaft bushings, worm saddles, M8 wheel bolt (head on the outer boss, nut in the inner face hex pocket), engaged and swung-out swing arm stop posts
launch_aim_lugpin_clip	launch	PLA	1	0.1	Printed C clip on the lug pin, outboard of the conrod
launch_carriage	launch	PLA	1	59.6	Carriage: rack, shoes, dart cradle, pusher with tether hole, spring arms and pegs, SW feet
launch_carriage_yoke	launch	PLA	1	23.5	Bolted yoke: both spring arms (hit the stop, d8.1 holes for the 8 mm steel rod spring pegs, D5 holes at y +-56 empty), cross bar, keys, SW_SEATED block, rigid SW_HOME foot. Prints on the peg boss bottoms and key plates (w 62)
launch_drive_washer	launch	PLA	1	0.2	Spacer washer: drive gear hub to the left cheek
launch_gear_clip	launch	PLA	1	0.3	C clip on the big gear pin groove
launch_gear_washer	launch	PLA	1	0.5	Spacer washer: big gear hub to the left cheek
launch_guard_l_a	launch	PLA	1	65.0	Spring guard left rear (end wall, top cover) (D33): 2 mm side plate outside both spring pairs, 4 wall brackets (M3x12 self tap into the rail wall); fit by moving it inward
launch_guard_l_b1	launch	PLA	1	4.5	Spring guard left mid front of the stop frame (D33): 2 mm side plate outside both spring pairs, 1 wall bracket (M3x12 self tap into the rail wall); fit by moving it inward
launch_guard_l_b2	launch	PLA	1	17.9	Spring guard left mid under the stop spring (D33): 2 mm side plate outside both spring pairs, 1 wall bracket (M3x12 self tap into the rail wall) and a foot on the front plate (M3x10 self tap down); fit by moving it inward

part	subsystem	material	qty	mass g	notes
launch_guard_l_c	launch	PLA	1	4.8	Spring guard left front (top cover) (D33): 2 mm side plate outside both spring pairs, 2 bosses on the front plate edge (M3x12 self tap along y); fit by moving it inward
launch_guard_r_a	launch	PLA	1	65.0	Spring guard right rear (end wall, top cover) (D33): 2 mm side plate outside both spring pairs, 4 wall brackets (M3x12 self tap into the rail wall); fit by moving it inward
launch_guard_r_b1	launch	PLA	1	4.5	Spring guard right mid front of the stop frame (D33): 2 mm side plate outside both spring pairs, 1 wall bracket (M3x12 self tap into the rail wall); fit by moving it inward
launch_guard_r_b2	launch	PLA	1	21.8	Spring guard right mid under the stop spring (D33): 2 mm side plate outside both spring pairs, 2 wall brackets (M3x12 self tap into the rail wall) and a foot on the front plate (M3x10 self tap down); fit by moving it inward
launch_guard_r_c	launch	PLA	1	4.8	Spring guard right front (top cover) (D33): 2 mm side plate outside both spring pairs, 2 bosses on the front plate edge (M3x12 self tap along y); fit by moving it inward
launch_muzzle_roller	launch	PLA	1	0.5	Lip roller at the carriage front: the tether bends down over it in tow and retrieval (groove r 6 for the 1 mm line), 693ZZ inside
launch_muzzle_washer_l	launch	PLA	1	0.0	Lip roller thrust washer (0.9 thick, flange on the cheek side) between the bearing inner race and the cheek: put both in the roller bore, then drop roller and washers into the fork slot
launch_muzzle_washer_r	launch	PLA	1	0.0	Lip roller thrust washer (0.9 thick, flange on the cheek side) between the bearing inner race and the cheek: put both in the roller bore, then drop roller and washers into the fork slot
launch_pulley_top	launch	PLA	1	0.5	Rail-rear tether pulley (D50, the line runs under it), groove r 6, 693ZZ inside
launch_pulley_top_washer_l	launch	PLA	1	0.1	Printed top-hat thrust washer between fork plate and 693ZZ inner race
launch_pulley_top_washer_r	launch	PLA	1	0.1	Printed top-hat thrust washer between fork plate and 693ZZ inner race
launch_rail_front	launch	PLA	1	137.7	Rail front piece: shoe track end, spring anchor posts with straps, muzzle tie
launch_rail_mid	launch	PLA	1	178.6	Rail mid piece: shoe track, cheeks on the sleeve, big gear pin bosses, lever seats, SW_GAP bracket pilots
launch_rail_rear	launch	PLA	1	135.9	Rail rear piece: shoe track, walls, tether fork and two pulleys behind the rear end
launch_rail_splice_l	launch	PLA	1	0.3	Top splice strap over the front rail joint (4 x M3 x 10 self tap)
launch_rail_splice_r	launch	PLA	1	0.3	Top splice strap over the front rail joint (4 x M3 x 10 self tap)
launch_stop_frame_l	launch	PLA	1	32.7	Stop frame (print outboard face down): outboard preload ledge, anchor seat, rear foot (2 x M3x12 into the rail wall), front foot (2 x M3x16 into the rail front plate, hex key through the d6.5 holes in the front wall). Fit before the spring guards and the main springs
launch_stop_frame_r	launch	PLA	1	32.7	Stop frame (print outboard face down): outboard preload ledge, anchor seat, rear foot (2 x M3x12 into the rail wall), front foot (2 x M3x16 into the rail front plate, hex key through the d6.5 holes in the front wall). Fit before the spring guards and the main springs

part	subsystem	material	qty	mass g	notes
launch_stop_spring_l	launch	PLA	1	81.6	Carriage stop flexure: 16 loops (31 tapered leaves t 3.15/1.0 x 14), print upside down on the flat w 83.5 face, 100% infill / all perimeters; preload 6.6 mm by 2 x M3x20 draw screws through the frame front wall
launch_stop_spring_r	launch	PLA	1	81.6	Carriage stop flexure: 16 loops (31 tapered leaves t 3.15/1.0 x 14), print upside down on the flat w 83.5 face, 100% infill / all perimeters; preload 6.6 mm by 2 x M3x20 draw screws through the frame front wall
reel_bar_cap_1	reel	PLA	1	1.0	Bar guide cap: closes the U guide over the bar. Print flat
reel_bar_cap_2	reel	PLA	1	1.0	Bar guide cap: closes the U guide over the bar. Print flat
reel_c_clip	reel	PLA	1	1.0	C-clip behind G_c
reel_cheek_i	reel	PLA	1	102.1	Inboard cheek: spool shaft 608, P2 bolts to its front edge. Print flat
reel_cheek_o	reel	PLA	1	102.1	Outboard cheek: spool shaft 608, P2 bolts to its front edge. Print flat
reel_cs_clip	reel	PLA	1	0.5	C-clip behind the C gear
reel_cup	reel	PLA	1	12.7	Wrap cup (Harry v7, ring hole turned axial): ring at the bottom at home, the line passes straight through it. Print open end down
reel_cup_clip	reel	PLA	1	0.3	Harry's v7 C-clip on the cup shaft neck in front of the cup
reel_k_clip	reel	PLA	1	0.1	C-clip holding the cup gear on the cup shaft
reel_k_clip2	reel	PLA	1	0.1	C-clip in front of the riser gear
reel_kc_clip	reel	PLA	1	0.1	C-clip behind P2
reel_p1_leg	reel	PLA	1	30.8	P1 leg: from the deck up to P1; lane T runs through its bushing. Print on its side
reel_p2_leg	reel	PLA	1	22.8	P2 leg: from the deck up to P2; lane T runs through its bushing. Print on its side
reel_post_t	reel	PLA	1	12.8	Lane T front post on the deck extension. Print on its foot
reel_s_clip	reel	PLA	1	0.1	C-clip holding the reel gear
reel_sleeve_clip	reel	PLA	1	1.0	C-clip on the sleeve behind the bus gear hub
reel_spacer_1	reel	PLA	1	2.9	Spacer tube between P1 and P2 (M3 through)
reel_spacer_2	reel	PLA	1	2.9	Spacer tube between P1 and P2 (M3 through)
reel_spacer_3	reel	PLA	1	2.9	Spacer tube between P1 and P2 (M3 through)
reel_spool	reel	PLA	1	36.1	Free storage spool: 20 m of 1 mm Dyneema on a dia 40 core, d62 flanges, keyed on the spool shaft
reel_t	reel	PLA	1	1.7	Thrust ring: the rocker pushes it onto the rear 608 inner race
reel_t_bracket	reel	PLA	1	58.3	T bracket: bushings for T3 and the spool shaft behind the 45T pair. Print flat
reel_tab_c	reel	PLA	1	6.6	Lane C bushing tab: screwed to the wall front. Print on its flange
reel_tab_k	reel	PLA	1	5.6	Lane K bushing tab: screwed to the wall front. Print on its flange
reel_wall	reel	PLA	1	291.1	Reel wall: foot, take-off bosses (C, K2, T2, idlers), rocker ears, stop pins, brake guides and crank boss. Print on its foot
show_aim_knob	show	PLA	1	4.6	Show build: hand aim knob on the aim countershaft end (D bore, a drop of glue): 10 turns per rail cycle 6 -> 47 -> 6 deg; the self-locking worm holds the pitch

part	subsystem	material	qty	mass g	notes
show_crank_handle	show	PLA	1	4.6	Show build: crank handle peg, pressed (or glued) into the arm hole. Prints grip end down
show_hand_crank	show	PLA	1	21.9	Show build: hand crank on the main shaft spigot (D bore, M3 onto the flat; its hub holds the shaft against the launch sleeve end). Turns the whole train. Prints arm face down
drum_axle	drum	PPA-CF	1	13.1	Fixed drum axle d10, pressed into brackets A and B
drum_collar	drum	PPA-CF	1	44.0	Dog collar: hex bore, 4 back-cut roofed dogs each face, fork groove (bridges over the lower dogs)
drum_coupler	drum	PPA-CF	1	89.3	Coupler: 5 N m shear pin limiter (D8), bus gear 50T m1.5 (D25), SW_INDEX cam, REEL journal, hex socket; prints hub end down, tree support under the bus gear ring
drum_coupler_hub	drum	PPA-CF	1	14.6	Coupler hub on the wiper shaft (bore parametric: WIPER), M8 nut inside, shear pin hole
drum_dogring_launch	drum	PPA-CF	1	19.0	LAUNCH dog ring: 3 slots for the launch sleeve lugs, 3 x M3 from the sleeve side
drum_gear_reel	drum	PPA-CF	1	95.9	REEL output gear m2 50T face 12 (D20), free on its sleeve, dog face toward the collar
drum_idler	drum	PPA-CF	1	7.6	Wrap idler shaft: pinion 10T m1.25 (locked at home by the drum lock arc), lock cam, two journals
drum_idler_gear	drum	PPA-CF	1	27.5	Wrap idler output gear 42T m1.5 (D bore) to the cad-reel cup ring 24T: 7 cup turns
drum_rail	drum	PPA-CF	1	13.1	Shift rail d10, pressed into brackets A and B
drum_reel_washer	drum	PPA-CF	1	0.9	REEL gear retaining washer on the coupler journal end, 2 x M3 x 6
drum_shaft	drum	PPA-CF	1	52.4	Main shaft: hex 22 AF, d12 spigot through the launch sleeve and on to the hand crank (show build); prints on a hex flat (support under the spigot)
frame_motor_adapter	frame	PPA-CF	1	299.3	puck adapter: slotted M6 holes (+-4 mm) for the motor bosses, rear lip with 8 M3 inserts for the rim screws; reprint when the motor is measured
launch_aim_bevel30	launch	PPA-CF	1	8.8	Bevel m1.5 z30 on the countershaft D flat, drives the z15 2:1
launch_aim_conrod	launch	PPA-CF	1	14.2	Conrod 156 between pin centres (PPA-CF), printed C clips on both pins
launch_aim_crank	launch	PPA-CF	1	35.3	Worm wheel m2 z20 face 16 = crank gear, crank arm r30 and d8 crank pin in one part, two 608 inside
launch_aim_lug	launch	PPA-CF	1	16.9	Lug bracket (PPA-CF), 2 x M3 into the rail wall, d8 lug pin at (u 115.2, w 0)
launch_aim_output	launch	PPA-CF	1	27.2	Splitter output m1.5 z30 face 12 with its d8 countershaft (D flat); show build: no splitter, the countershaft runs on out of the housing to the hand aim knob
launch_aim_worm	launch	PPA-CF	1	21.3	Worm m2 pd24 single start (lead angle 4.8 deg, self-locking) + d8 shaft + bevel z15
launch_big_gear	launch	PPA-CF	1	177.6	Big gear m2.5: sector layer 27 of 40 teeth (rack, v10 release teeth) + full 40T layer (drive), SW_GAP cam bump, on the rail pin
launch_drive_gear	launch	PPA-CF	1	29.9	Drive gear m2.5 26T, sprag #1 inside on an integral key rib
launch_drive_hub	launch	PPA-CF	1	22.0	Show build: printed fixed hub in place of sprag #1 (20 x 47 x 14), keyed to the sleeve rib and the drive gear rib, so the hand crank cocks in both senses
launch_gear_pin	launch	PPA-CF	1	5.4	Snap pin d10 for the big gear, C clip; prints lying on its flat

part	subsystem	material	qty	mass g	notes
launch_sleeve	launch	PPA-CF	1	49.3	Launch sleeve d20: dog ring flange, integral key ribs (sprag #1, splitter sun), journals
launch_yoke_peg_l	launch	PPA-CF	1	1.2	Spring peg on the yoke arm (printed PPA-CF lying down, or 8 mm steel rod 20 long): pressed into the blind d8.1 hole from below up to its end (a drop of CA if it is loose), the spring hook sits on the 6 mm below the boss (held at w 60 by the spring line); ends deburred
launch_yoke_peg_r	launch	PPA-CF	1	1.2	Spring peg on the yoke arm (printed PPA-CF lying down, or 8 mm steel rod 20 long): pressed into the blind d8.1 hole from below up to its end (a drop of CA if it is loose), the spring hook sits on the 6 mm below the boss (held at w 60 by the spring line); ends deburred
reel_bar	reel	PPA-CF	1	7.9	Brake bar: cad-drum's track 2 cam pushes its nose, its left end turns the crank. Print flat
reel_bc	reel	PPA-CF	1	7.5	C bevel m2 z20 (1:1 into lane C)
reel_bcx	reel	PPA-CF	1	8.0	Lane C bevel m2 z20 (meshes the C bevel)
reel_bk2	reel	PPA-CF	1	3.3	K bevel m1.5 z18 on K2 (1:1 into lane K)
reel_bkx	reel	PPA-CF	1	3.1	Lane K bevel m1.5 z18 (meshes the K2 bevel)
reel_bt2	reel	PPA-CF	1	1.5	T bevel m1 z20 on T2 (1:1 into lane T)
reel_bt3	reel	PPA-CF	1	1.3	T3 bevel m1 z20
reel_bt3x	reel	PPA-CF	1	1.4	Lane T rear bevel m1 z20 (meshes T3 in front of its axis)
reel_btx	reel	PPA-CF	1	1.4	Lane T front bevel m1 z20 (meshes the T2 bevel)
reel_bus	reel	PPA-CF	1	26.3	Bus gear m1.5 z46 (free on the sleeve): driven by the drum's bus gear, drives the swing idler. Print face down
reel_c	reel	PPA-CF	1	19.8	Capstan tube C: key rib in the sprag and the drum, D flat for G_c. Print standing
reel_crank	reel	PPA-CF	1	2.1	Brake bell crank: the bar turns it, arm B pushes the rocker leg. Print on its flat side
reel_crank_washer	reel	PPA-CF	1	0.2	Printed washer: keeps the crank on its boss
reel_cs	reel	PPA-CF	1	9.3	Countershaft C (PPA-CF d16): bevel in front of the wall, wall bushing, 22T gear behind
reel_drum	reel	PPA-CF	1	44.5	Capstan drum dia 40 x 40 (Harry v7), keyed on the tube. Print flange down
reel_ga	reel	PPA-CF	1	11.2	G_a m2 z20 on lane C
reel_ga2	reel	PPA-CF	1	4.0	G_a' m1.5 z20 on lane K
reel_gc	reel	PPA-CF	1	19.4	C gear m2 z22: driven by the REEL gear (50T) through the 41T idler
reel_gc2	reel	PPA-CF	1	9.2	G_c m2 z20 on the capstan tube
reel_gk	reel	PPA-CF	1	11.9	Cup gear m1.5 z24: driven by the drum's wrap idler; the raised track on its back holds SW_ARM closed, the notch at home lets it open
reel_gk1	reel	PPA-CF	1	9.3	Cup riser gear m1.5 z30 on the cup shaft in front of the wall (1:1 riser to K2)
reel_gk2	reel	PPA-CF	1	9.5	K2 riser gear m1.5 z30
reel_gkc	reel	PPA-CF	1	3.8	G_k m1.5 z20 on the cup shaft K'
reel_gr	reel	PPA-CF	1	4.7	Reel gear m1.5 z20: turns the clutch disk 2.5 x wiper speed
reel_gs	reel	PPA-CF	1	2.9	Spool shaft gear m1.5 z16
reel_gt2	reel	PPA-CF	1	9.8	T2 gear m1.5 z28: meshes the clutch hub 1:1

part	subsystem	material	qty	mass g	notes
reel_gt3	reel	PPA-CF	1	2.9	T3 gear m1.5 z16
reel_hub	reel	PPA-CF	1	13.1	Clutch hub: two 608s on the reel shaft, 40T teeth drive T2 (and the spool 1:1). Print rear face down
reel_hub_flange	reel	PPA-CF	1	6.0	Friction disc: keeps the front 608 in, carries the clutch friction ring. Print flat
reel_ic	reel	PPA-CF	1	67.6	REEL idler m2 z41: from the REEL gear to the C gear
reel_ic_axle	reel	PPA-CF	1	2.0	REEL idler axle pin (PPA-CF d8): pressed into the wall from behind, head behind the idler
reel_icr	reel	PPA-CF	1	105.6	I_c m2 z58: idler from lane C to the capstan; show build: plain bore, turns free on its axle (no sprag #2)
reel_icr_axle	reel	PPA-CF	1	11.7	I_c axle (PPA-CF d20, plain in the show build): fixed in P1 (D flat), the I_c idler turns on it
reel_ik1	reel	PPA-CF	1	12.3	Cup riser idler m1.5 z34 on its axle pin in the wall
reel_ik1_axle	reel	PPA-CF	1	1.2	Idler axle pin (PPA-CF d8): pressed into the wall from the front, head in front of the idler
reel_ik2	reel	PPA-CF	1	9.4	Cup riser idler m1.5 z30 on its axle pin in the wall
reel_ik2_axle	reel	PPA-CF	1	1.2	Idler axle pin (PPA-CF d8): pressed into the wall from the front, head in front of the idler
reel_ikr	reel	PPA-CF	1	63.1	I_k m1.5 z76: idler from lane K to the cup shaft
reel_ikr_axle	reel	PPA-CF	1	0.6	I_k axle pin (PPA-CF d8): pressed into P2, head in front of the idler
reel_k	reel	PPA-CF	1	5.5	Cup shaft K: through the sleeve, cup gear behind the wall, riser gear in front. Print standing
reel_k2	reel	PPA-CF	1	2.0	K2 stub (PPA-CF d8): wall bushing, riser gear and the K bevel
reel_kc	reel	PPA-CF	1	9.6	Rear cup shaft K': cup in front, G_k behind, journal in P2. Print standing
reel_lane_c	reel	PPA-CF	1	21.5	Lane C (PPA-CF d12): the capstan drive along the deck to the reel unit. Print standing
reel_lane_k	reel	PPA-CF	1	14.5	Lane K (PPA-CF d8): the cup drive along the deck to the reel unit. Print standing
reel_lane_t	reel	PPA-CF	1	12.6	Lane T (PPA-CF d8): the take-up drive along the deck to the spool. Print standing
reel_p1	reel	PPA-CF	1	111.9	Reel unit front plate P1 with the sprag #2 seat and the capstan bushing. Print front face down
reel_p2	reel	PPA-CF	1	86.0	Reel unit rear plate P2: cup shaft and lane K journals, I_k axle. Print flat
reel_rocker	reel	PPA-CF	1	6.1	Brake rocker: flexible PPA-CF lever = clutch spring, pushed by the crank. Print flat
reel_s	reel	PPA-CF	1	14.1	Reel shaft: 608 journals, head = clutch disk, D flat for the reel gear. Print head down
reel_sleeve	reel	PPA-CF	1	8.4	Stationary sleeve at the old capstan axis: c1 turns on it, the cup shaft inside. Print flange down
reel_ss	reel	PPA-CF	1	9.6	Spool shaft (PPA-CF d8): 45T gear, bushing in the T bracket, 608s in both cheeks, the spool on its flat
reel_t2	reel	PPA-CF	1	2.4	T2 stub (PPA-CF d8): wall bushing, 40T gear and the T bevel
reel_t3	reel	PPA-CF	1	1.2	T3 stub (PPA-CF d8): rear T bevel and the 45T gear
reel_ti	reel	PPA-CF	1	14.7	Take-up idler m1.5 z34

part	subsystem	material	qty	mass g	notes
reel_ti_axle	reel	PPA-CF	1	1.0	Take-up idler axle pin (PPA-CF d8): pressed into the T bracket, head on the inboard side of the idler

Printed mass: PLA 4073 g (1365.7 m of filament); PPA-CF 1938 g (644.6 m of filament).

Bought parts

item	type	count	used by
608ZZ 8x22x7	bearings	6	launch_aim_brg_1, launch_aim_brg_2, reel_brg_1, reel_brg_2, reel_brg_3, reel_brg_4
693ZZ 3x8x4	bearings	2	launch_muzzle_roller_brg, launch_pulley_top_brg
M3 nut	fasteners	5	reel_p1
M3 nyloc nut	fasteners	1	launch_muzzle_roller_axle
M3x10 self tap	fasteners	2	launch_guard_l_b2, launch_guard_r_b2
M3x10 SHCS	fasteners	18	reel_tab_c, reel_tab_k, reel_post_t, reel_p1_leg, reel_p2_leg, reel_cheek_i, reel_cheek_o, reel_t_bracket
M3x10 SHCS + M3 nut	fasteners	1	reel_firm_screw
M3x12 self tap	fasteners	17	launch_guard_l_a, launch_guard_l_b1, launch_guard_l_b2, launch_guard_l_c, launch_guard_r_a, launch_guard_r_b1, launch_guard_r_b2, launch_guard_r_c
M3x12 SHCS	fasteners	2	reel_p2
M3x16 SHCS	fasteners	15	launch_muzzle_roller_axle, reel_bar_screws, reel_p1, reel_p2
M3x20 SHCS	fasteners	1	reel_crank_screw
M3x20 SHCS + nyloc nut	fasteners	1	launch_pulley_top_axle
M3x25 SHCS + M3 nut	fasteners	1	reel_rocker_pin
M3x6 set screw or SHCS	fasteners	1	show_hand_crank
M3x8 SHCS	fasteners	5	reel_sleeve, reel_trim_screws
M8 hex nut ISO 4032	fasteners	1	launch_aim_nut
M8x25 hex bolt ISO 4017	fasteners	1	launch_aim_bolt
(part of the wiper motor)	other	1	frame_wiper_shaft
12 V car wiper motor (self-locking worm), 2-wire rebuild	other	1	frame_wiper_motor
8 mm steel rod, cut to length (see note), ends deburred	other	2	launch_anchor_pin_l, launch_anchor_pin_r
Century C-243 (5/8 x 6-1/2 in x 0.054 in wire) extension spring	other	2	launch_spring_l_hook_a, launch_spring_r_hook_a
Dyneema braid 1 mm (LINE)	other	9	launch_line_a0, launch_line_a1, launch_line_a2, launch_line_a3, launch_line_a4, launch_line_a5, launch_line_a6, launch_line_a7, launch_line_a8
KEDGE v34 dart (reference, built separately)	other	1	launch_dart
launch_line_p (bought, no bought_ref)	other	1	launch_line_p
launch_spring_l_0 (bought, no bought_ref)	other	1	launch_spring_l_0
launch_spring_l_1 (bought, no bought_ref)	other	1	launch_spring_l_1
launch_spring_l_2 (bought, no bought_ref)	other	1	launch_spring_l_2
launch_spring_l_3 (bought, no bought_ref)	other	1	launch_spring_l_3

item	type	count	used by
launch_spring_l_4 (bought, no bought_ref)	other	1	launch_spring_l_4
launch_spring_l_hook_c (bought, no bought_ref)	other	1	launch_spring_l_hook_c
launch_spring_r_0 (bought, no bought_ref)	other	1	launch_spring_r_0
launch_spring_r_1 (bought, no bought_ref)	other	1	launch_spring_r_1
launch_spring_r_2 (bought, no bought_ref)	other	1	launch_spring_r_2
launch_spring_r_3 (bought, no bought_ref)	other	1	launch_spring_r_3
launch_spring_r_4 (bought, no bought_ref)	other	1	launch_spring_r_4
launch_spring_r_hook_c (bought, no bought_ref)	other	1	launch_spring_r_hook_c
N20 right-angle self-locking worm gearmotor 12 V 27 rpm (have)	other	1	drum_n20

4. Print list

160 printed parts on 20 plates. The 3MF plates and STL files are in the print pack.

part	material	qty	print size mm	supports
frame_deck_4	PLA	1	229 x 246 x 8	no
frame_deck_2	PLA	1	220 x 246 x 8	no
frame_deck_ext	PLA	1	247 x 208 x 8	no
reel_wall	PLA	1	201 x 82 x 213	yes, tree supports from the plate (1757 mm2 over 45 deg)
frame_cheek_r	PLA	1	215 x 220 x 30	yes, tree supports from the plate (5117 mm2 over 45 deg)
frame_deck_3	PLA	1	229 x 134 x 8	no
frame_deck_1	PLA	1	220 x 134 x 8	no
launch_aim_housing	PLA	1	176 x 59 x 183	yes, tree supports from the plate (761 mm2 over 45 deg)
launch_rail_mid	PLA	1	92 x 68 x 170	yes, tree supports from the plate (2315 mm2 over 45 deg)
drum_drum_a	PLA	1	97 x 97 x 75	yes, tree supports from the plate (732 mm2 over 45 deg)
frame_motor_leg	PLA	1	49 x 69 x 187	yes, tree supports from the plate (222 mm2 over 45 deg)
launch_rail_rear	PLA	1	181 x 68 x 76	yes, tree supports from the plate (4407 mm2 over 45 deg)
launch_rail_front	PLA	1	132 x 128 x 48	yes, tree supports from the plate (7890 mm2 over 45 deg)
reel_cheek_i	PLA	1	63 x 226 x 22	yes, tree supports from the plate (13980 mm2 over 45 deg)
reel_cheek_o	PLA	1	63 x 226 x 22	yes, tree supports from the plate (375 mm2 over 45 deg)
drum_drum_b	PLA	1	60 x 60 x 58	yes, tree supports from the plate (1818 mm2 over 45 deg)
launch_stop_spring_l	PLA	1	155 x 52 x 14	no

part	material	qty	print size mm	supports
launch_stop_spring_r	PLA	1	155 x 52 x 14	no
reel_t_bracket	PLA	1	48 x 196 x 20	no
launch_guard_l_a	PLA	1	250 x 49 x 40	no
launch_guard_r_a	PLA	1	250 x 49 x 40	no
launch_carriage	PLA	1	205 x 240 x 43	yes, tree supports from the plate (6453 mm2 over 45 deg)
drum_bracket_a	PLA	1	105 x 117 x 7	no
frame_guard_front	PLA	1	155 x 104 x 16	no
reel_spool	PLA	1	62 x 62 x 38	yes, tree supports from the plate (4580 mm2 over 45 deg)
frame_can_stay	PLA	1	55 x 55 x 71	yes, tree supports from the plate (906 mm2 over 45 deg)
reel_p1_leg	PLA	1	26 x 36 x 125	no
launch_stop_frame_l	PLA	1	175 x 53 x 55	yes, tree supports from the plate (541 mm2 over 45 deg)
launch_stop_frame_r	PLA	1	175 x 53 x 55	yes, tree supports from the plate (541 mm2 over 45 deg)
drum_bracket_b	PLA	1	79 x 117 x 6	no
reel_p2_leg	PLA	1	26 x 26 x 125	no
frame_guard_hood	PLA	1	122 x 60 x 14	no
show_hand_crank	PLA	1	70 x 22 x 59	no
drum_fork	PLA	1	87 x 94 x 83	yes, tree supports from the plate (1302 mm2 over 45 deg)
launch_carriage_yoke	PLA	1	43 x 127 x 28	yes, tree supports from the plate (1406 mm2 over 45 deg)
launch_guard_r_b2	PLA	1	145 x 32 x 40	no
launch_guard_l_b2	PLA	1	145 x 32 x 40	no
drum_bracket_idler2	PLA	1	50 x 141 x 6	no
drum_bracket_idler	PLA	1	52 x 71 x 7	no
reel_post_t	PLA	1	26 x 14 x 120	no
reel_cup	PLA	1	48 x 53 x 18	yes, tree supports from the plate (1210 mm2 over 45 deg)
drum_n20_cradle	PLA	1	32 x 54 x 19	yes, tree supports from the plate (108 mm2 over 45 deg)
reel_tab_c	PLA	1	16 x 26 x 34	no
show_crank_handle	PLA	1	14 x 14 x 38	no
reel_tab_k	PLA	1	16 x 20 x 36	no
launch_aim_cap	PLA	1	46 x 14 x 19	no
show_aim_knob	PLA	1	28 x 28 x 8	no
launch_guard_l_c	PLA	1	32 x 54 x 26	no
launch_guard_r_c	PLA	1	32 x 54 x 26	no
launch_guard_l_b1	PLA	1	12 x 49 x 40	no
launch_guard_r_b1	PLA	1	12 x 49 x 40	no
drum_hand_lever	PLA	1	46 x 24 x 6	no
reel_spacer_1	PLA	1	10 x 10 x 34	no

part	material	qty	print size mm	supports
reel_spacer_2	PLA	1	10 x 10 x 34	no
reel_spacer_3	PLA	1	10 x 10 x 34	no
drum_n20_strap	PLA	1	18 x 54 x 3	no
reel_t	PLA	1	26 x 26 x 5	no
reel_bar_cap_1	PLA	1	10 x 20 x 5	no
reel_bar_cap_2	PLA	1	10 x 20 x 5	no
reel_sleeve_clip	PLA	1	27 x 25 x 3	no
reel_c_clip	PLA	1	27 x 25 x 3	no
launch_muzzle_roller	PLA	1	14 x 14 x 5	no
launch_pulley_top	PLA	1	14 x 14 x 5	no
reel_cs_clip	PLA	1	22 x 20 x 2	no
launch_gear_washer	PLA	1	18 x 18 x 2	no
launch_gear_clip	PLA	1	17 x 18 x 1	no
reel_cup_clip	PLA	1	14 x 14 x 2	no
launch_rail_splice_l	PLA	1	22 x 5 x 3	no
launch_rail_splice_r	PLA	1	22 x 5 x 3	no
launch_drive_washer	PLA	1	32 x 32 x 1	no
reel_k_clip	PLA	1	14 x 13 x 1	no
reel_k_clip2	PLA	1	14 x 13 x 1	no
reel_kc_clip	PLA	1	14 x 13 x 1	no
launch_pulley_top_washer_l	PLA	1	9 x 9 x 2	no
launch_pulley_top_washer_r	PLA	1	9 x 9 x 2	no
launch_aim_crankpin_clip	PLA	1	11 x 10 x 1	no
launch_aim_lugpin_clip	PLA	1	11 x 10 x 1	no
reel_s_clip	PLA	1	11 x 10 x 1	no
launch_muzzle_washer_l	PLA	1	6 x 6 x 1	no
launch_muzzle_washer_r	PLA	1	6 x 6 x 1	no
frame_motor_adapter	PPA-CF	1	194 x 194 x 16	yes, tree supports from the plate (6735 mm2 over 45 deg)
launch_big_gear	PPA-CF	1	105 x 105 x 27	yes, tree supports from the plate (639 mm2 over 45 deg)
reel_icr	PPA-CF	1	120 x 120 x 14	yes, tree supports from the plate (8054 mm2 over 45 deg)
reel_p1	PPA-CF	1	161 x 185 x 14	no
drum_gear_reel	PPA-CF	1	104 x 104 x 19	no
drum_coupler	PPA-CF	1	78 x 78 x 62	yes, tree supports from the plate (1998 mm2 over 45 deg)
reel_p2	PPA-CF	1	155 x 148 x 6	no
reel_ic	PPA-CF	1	86 x 86 x 12	no
reel_ikr	PPA-CF	1	117 x 117 x 5	no
drum_shaft	PPA-CF	1	227 x 25 x 22	yes, tree supports from the plate (1465 mm2 over 45 deg)
reel_drum	PPA-CF	1	46 x 46 x 43	yes, tree supports from the plate (251 mm2 over 45 deg)

part	material	qty	print size mm	supports
launch_sleeve	PPA-CF	1	64 x 64 x 85	yes, tree supports from the plate (2907 mm2 over 45 deg)
drum_collar	PPA-CF	1	60 x 60 x 26	yes, tree supports from the plate (2694 mm2 over 45 deg)
launch_aim_crank	PPA-CF	1	62 x 60 x 38	yes, tree supports from the plate (985 mm2 over 45 deg)
launch_drive_gear	PPA-CF	1	70 x 70 x 29	yes, tree supports from the plate (2457 mm2 over 45 deg)
launch_aim_output	PPA-CF	1	48 x 48 x 74	no
drum_idler_gear	PPA-CF	1	66 x 66 x 9	no
reel_bus	PPA-CF	1	72 x 72 x 10	no
launch_drive_hub	PPA-CF	1	47 x 47 x 14	no
reel_lane_c	PPA-CF	1	12 x 12 x 161	no
launch_aim_worm	PPA-CF	1	28 x 28 x 54	yes, tree supports from the plate (2779 mm2 over 45 deg)
reel_c	PPA-CF	1	28 x 28 x 65	yes, tree supports from the plate (391 mm2 over 45 deg)
reel_gc	PPA-CF	1	48 x 48 x 12	no
drum_dogring_launch	PPA-CF	1	62 x 62 x 10	yes, tree supports from the plate (123 mm2 over 45 deg)
launch_aim_lug	PPA-CF	1	30 x 58 x 28	yes, tree supports from the plate (995 mm2 over 45 deg)
reel_ti	PPA-CF	1	54 x 54 x 6	no
drum_coupler_hub	PPA-CF	1	26 x 26 x 32	no
reel_lane_k	PPA-CF	1	8 x 8 x 234	no
launch_aim_conrod	PPA-CF	1	173 x 17 x 6	no
reel_s	PPA-CF	1	62 x 62 x 48	no
reel_hub	PPA-CF	1	45 x 45 x 17	yes, tree supports from the plate (778 mm2 over 45 deg)
drum_axle	PPA-CF	1	141 x 10 x 9	no
drum_rail	PPA-CF	1	141 x 10 x 9	no
reel_icr_axle	PPA-CF	1	26 x 26 x 33	yes, tree supports from the plate (315 mm2 over 45 deg)
reel_lane_t	PPA-CF	1	8 x 8 x 203	no
reel_ik1	PPA-CF	1	54 x 54 x 5	no
reel_gk	PPA-CF	1	39 x 39 x 23	yes, tree supports from the plate (668 mm2 over 45 deg)
reel_ga	PPA-CF	1	44 x 44 x 8	no
reel_cs	PPA-CF	1	16 x 16 x 43	no
reel_gt2	PPA-CF	1	45 x 45 x 6	no
reel_ss	PPA-CF	1	8 x 8 x 157	no
reel_kc	PPA-CF	1	10 x 10 x 103	no
reel_gk2	PPA-CF	1	48 x 48 x 5	no
reel_ik2	PPA-CF	1	48 x 48 x 5	no
reel_gk1	PPA-CF	1	48 x 48 x 5	no
reel_gc2	PPA-CF	1	44 x 44 x 8	no

part	material	qty	print size mm	supports
launch_aim_bevel30	PPA-CF	1	53 x 52 x 10	yes, tree supports from the plate (590 mm2 over 45 deg)
reel_sleeve	PPA-CF	1	34 x 34 x 24	no
reel_bcx	PPA-CF	1	44 x 44 x 8	yes, tree supports from the plate (337 mm2 over 45 deg)
reel_bar	PPA-CF	1	154 x 19 x 16	no
drum_idler	PPA-CF	1	22 x 22 x 46	yes, tree supports from the plate (331 mm2 over 45 deg)
reel_bc	PPA-CF	1	44 x 44 x 8	yes, tree supports from the plate (337 mm2 over 45 deg)
reel_rocker	PPA-CF	1	29 x 92 x 6	no
reel_hub_flange	PPA-CF	1	62 x 62 x 2	no
reel_k	PPA-CF	1	10 x 10 x 59	no
launch_gear_pin	PPA-CF	1	53 x 16 x 12	no
reel_gr	PPA-CF	1	33 x 33 x 6	no
reel_ga2	PPA-CF	1	33 x 33 x 5	no
reel_gkc	PPA-CF	1	33 x 33 x 5	no
reel_bk2	PPA-CF	1	30 x 30 x 8	yes, tree supports from the plate (166 mm2 over 45 deg)
reel_bkx	PPA-CF	1	30 x 30 x 7	yes, tree supports from the plate (166 mm2 over 45 deg)
reel_gt3	PPA-CF	1	27 x 27 x 6	no
reel_gs	PPA-CF	1	27 x 27 x 6	no
reel_t2	PPA-CF	1	8 x 8 x 39	no
reel_crank	PPA-CF	1	21 x 28 x 12	no
reel_ic_axle	PPA-CF	1	12 x 12 x 31	no
reel_k2	PPA-CF	1	8 x 8 x 33	no
reel_bt2	PPA-CF	1	22 x 22 x 7	no
reel_btx	PPA-CF	1	22 x 22 x 6	no
reel_bt3x	PPA-CF	1	22 x 22 x 6	no
reel_bt3	PPA-CF	1	22 x 22 x 6	no
reel_ik1_axle	PPA-CF	1	12 x 12 x 18	no
reel_ik2_axle	PPA-CF	1	12 x 12 x 18	no
reel_t3	PPA-CF	1	8 x 8 x 21	no
launch_yoke_peg_l	PPA-CF	1	20 x 8 x 7	no
launch_yoke_peg_r	PPA-CF	1	20 x 8 x 7	no
reel_ti_axle	PPA-CF	1	12 x 12 x 15	no
drum_reel_washer	PPA-CF	1	35 x 35 x 1	no
reel_ikr_axle	PPA-CF	1	12 x 12 x 8	no
reel_crank_washer	PPA-CF	1	12 x 12 x 1	no

plate_1_PLA.3mf: launch_carriage, launch_drive_washer, launch_guard_l_c, launch_guard_r_b2.

plate_2_PLA.3mf: frame_cheek_r, launch_muzzle_roller, launch_pulley_top, reel_c_clip, reel_cs_clip, reel_cup_clip, reel_p1_leg, reel_p2_leg, reel_post_t, reel_t, reel_tab_c, reel_tab_k, show_crank_handle, show_hand_crank.

plate_3_PLA.3mf: drum_hand_lever, frame_deck_3, launch_aim_crankpin_clip, launch_aim_lugpin_clip, launch_guard_l_b1, launch_guard_r_b1, launch_guard_r_c, launch_rail_splice_l, reel_bar_cap_2, reel_wall.

plate_4_PLA.3mf: drum_n20_cradle, drum_n20_strap, frame_deck_1, frame_guard_front, launch_gear_clip, launch_gear_washer, reel_sleeve_clip, reel_spool.

plate_5_PLA.3mf: drum_bracket_a, launch_aim_cap, launch_guard_l_b2, launch_rail_front, reel_cheek_i.

plate_6_PLA.3mf: drum_drum_a, launch_aim_housing, launch_muzzle_washer_l, launch_muzzle_washer_r, launch_rail_mid, launch_rail_rear, launch_rail_splice_r, reel_cheek_o.

plate_7_PLA.3mf: drum_bracket_b, drum_fork, launch_stop_frame_l, launch_stop_frame_r, reel_k_clip, reel_k_clip2, reel_kc_clip, reel_s_clip, reel_spacer_1, reel_spacer_2, reel_spacer_3, reel_t_bracket.

plate_8_PLA.3mf: drum_bracket_idler, drum_bracket_idler2, drum_drum_b, frame_guard_hood, launch_carriage_yoke, launch_pulley_top_washer_l, launch_pulley_top_washer_r, launch_stop_spring_l, launch_stop_spring_r, show_aim_knob.

plate_9_PLA.3mf: frame_can_stay, frame_motor_leg, reel_bar_cap_1, reel_cup.

plate_10_PLA.3mf: frame_deck_4.

plate_11_PLA.3mf: frame_deck_2.

plate_12_PLA.3mf: frame_deck_ext.

plate_13_PLA.3mf: launch_guard_l_a.

plate_14_PLA.3mf: launch_guard_r_a.

plate_15_PPA-CF.3mf: drum_shaft, frame_motor_adapter, launch_drive_hub, launch_gear_pin, reel_cs, reel_gk, reel_gt2, reel_hub, reel_ik1_axle, reel_ik2_axle.

plate_16_PPA-CF.3mf: drum_coupler, launch_aim_lug, launch_drive_gear, reel_bar, reel_bus, reel_p1, reel_rocker.

plate_17_PPA-CF.3mf: drum_dogring_launch, drum_idler_gear, drum_reel_washer, launch_aim_conrod, launch_sleeve, reel_crank, reel_ic, reel_k, reel_p2, reel_s.

plate_18_PPA-CF.3mf: drum_axle, drum_gear_reel, launch_big_gear, reel_crank_washer, reel_icr, reel_ikr, reel_ikr_axle, reel_ti_axle.

plate_19_PPA-CF.3mf: drum_collar, drum_rail, launch_aim_bevel30, launch_aim_crank, launch_aim_output, launch_yoke_peg_l, launch_yoke_peg_r, reel_bc, reel_bcx, reel_drum, reel_ga, reel_gc, reel_gc2, reel_gk1, reel_gk2, reel_hub_flange, reel_ic_axle, reel_ik1, reel_ik2, reel_k2, reel_kc, reel_lane_c, reel_lane_k, reel_lane_t, reel_ss, reel_t2, reel_t3, reel_ti.

plate_20_PPA-CF.3mf: drum_coupler_hub, drum_idler, launch_aim_worm, reel_bk2, reel_bkx, reel_bt2, reel_bt3, reel_bt3x, reel_btx, reel_c, reel_ga2, reel_gkc, reel_gr, reel_gs, reel_gt3, reel_icr_axle, reel_sleeve.

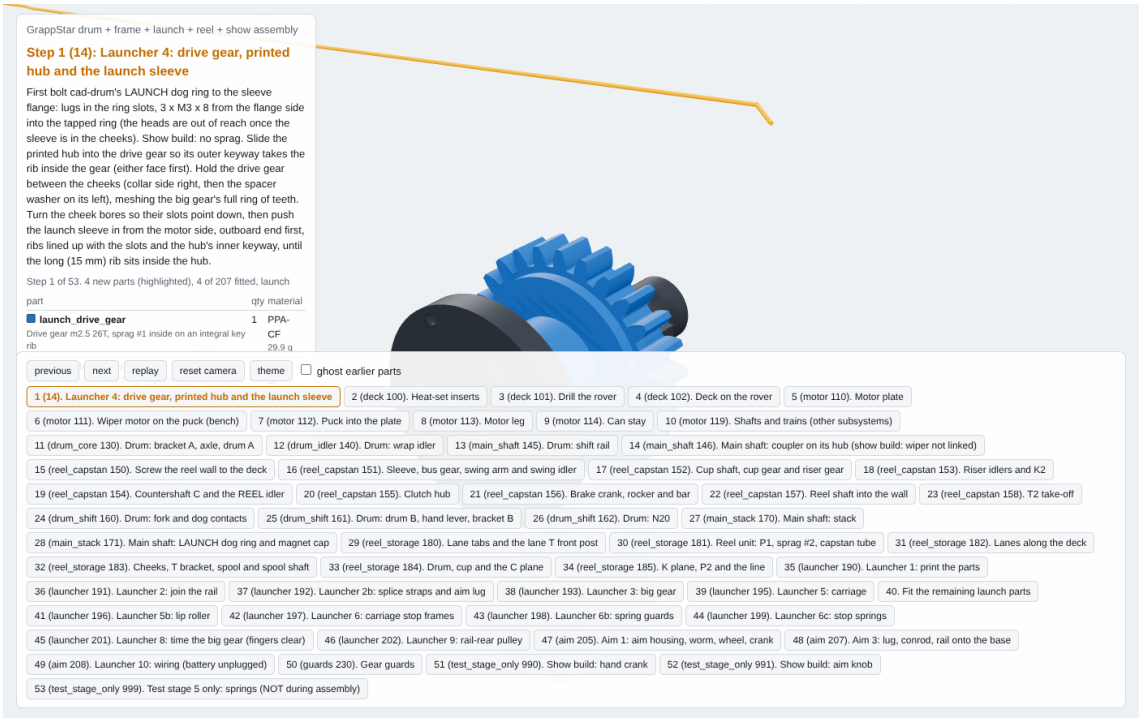
Print settings are in PRINT_SHEET.pdf. Use a hardened nozzle for PPA-CF and dry it first.

5. Assembly steps

Step 1 (14). Launcher 4: drive gear, printed hub and the launch sleeve

First bolt cad-drum's LAUNCH dog ring to the sleeve flange: lugs in the ring slots, 3 x M3 x 8 from the flange side into the tapped ring (the heads are out of reach once the sleeve is in the cheeks). Show build: no sprag. Slide the printed hub into the drive gear so its outer keyway takes the rib inside the gear (either face first). Hold the drive gear between the cheeks (collar side right, then the spacer washer on its left), meshing the big gear's full ring of teeth. Turn the cheek bores so their slots point down, then push the launch sleeve in from the motor side, outboard end first, ribs lined up with the slots and the hub's inner keyway, until the long (15 mm) rib sits inside the hub.

Parts: launch_drive_gear, launch_drive_hub, launch_sleeve, launch_drive_washer.



Step 2 (deck 100). Heat-set inserts

Battery unplugged. Soldering iron with an insert tip, about 220 C for PLA and about 280 C for the PPA-CF puck. Press every M3 x 4 x 5 insert square into its 4.0 mm hole until flush: deck (55), motor plate (2 for the leg bridge, 2 for the can stay, 3 for the gear hood), puck adapter (8 in the rear lip), bay (corner bosses, standoffs, clamp posts, perfboard bosses). Let each cool before the next one nearby.

Parts: frame_deck_1, frame_deck_2, frame_deck_3, frame_deck_4, frame_deck_ext, frame_cheek_r, frame_motor_adapter.

Step 3 (deck 101). Drill the rover

Lay the deck on the rover top plate where it will live (launch direction forward). Mark through the 54 perimeter holes with a 3 mm drill or a scriber (or tape down the 1:1 PDF template from cad/parts/_frame_tests/drill_template.py), take the deck off, drill 3.4 mm. Deburr.

Also handle (fitted earlier): frame_deck_1, frame_deck_2, frame_deck_3, frame_deck_4, frame_deck_ext.

Step 4 (deck 102). Deck on the rover

Slide the tiles together on the dovetails (from above). M3 x 25 SHCS down through every perimeter hole, washer and nyloc under the rover plate. Snug, not crushed. Do this before the other subsystems put their feet down: some rover bolts sit under their feet.

Also handle (fitted earlier): frame_deck_1, frame_deck_2, frame_deck_3, frame_deck_4, frame_deck_ext.

Step 5 (motor 110). Motor plate

Hold the motor plate upright (front face, the side without the bridge, towards +y) with its bridge against the top of the leg. 2 x M3 x 12 along y through the leg's long counterbores into the bridge inserts.

Also handle (fitted earlier): frame_cheek_r.

Step 6 (motor 111). Wiper motor on the puck (bench)

Battery unplugged. Put the M6 bolts through the puck slots from the front (heads sit in the slot counterbores), into the motor bosses (threaded bosses, or a nut behind each boss ear). Slide the motor so its shaft is centred in the bore (a 10 mm drill shank in the coupler bore helps), then tighten.

Parts: frame_wiper_motor, frame_wiper_shaft.
Also handle (fitted earlier): frame_motor_adapter.

Step 7 (motor 112). Puck into the plate

Feed the puck into the plate window from behind (the motor side) until its lip sits on the plate's rear face. From the front, M3 x 12 through the plate into the lip inserts (7 of the 8 rim holes have both plate and lip; use those).

Also handle (fitted earlier): frame_motor_adapter.

Step 8 (motor 113). Motor leg

Stand the leg on the deck in the corner between the reel and the bay, foot along y. 4 x M3 x 10 down through the foot into the deck inserts.

Parts: frame_motor_leg.

Step 9 (motor 114). Can stay

Post against the plate's rear face, 2 x M3 x 12 into the plate inserts. Cable tie or hose clamp round the motor can through the two cradle slots, firm but not crushing.

Parts: frame_can_stay.

Step 10 (motor 119). Shafts and trains (other subsystems)

Now cad-drum (coupler on the wiper shaft, SW_INDEX block on the strut of idler post 2: drum step "Main shaft: index contact"), cad-reel and cad-launch install their trains and aim housing. Follow their steps, then come back.

Step 11 (drum_core 130). Drum: bracket A, axle, drum A

Bolt bracket A to the deck (2 x M3 x 12). Press the drum axle into it (flat facing forward). Slide drum A onto the axle, gear first, and turn it so the gap in its lock ring faces back toward the idler position (about drum 180 on the pointer). The shift rail goes in after the idler.

Parts: drum_bracket_a, drum_axle, drum_drum_a.

Step 12 (drum_idler 140). Drum: wrap idler

Push the idler output gear onto the idler shaft flat from the long end. Slide idler post 2 onto the short journal and idler post 1 onto the long journal (the thick middle of the shaft sits between the posts and locates it). Fit this before cad-reel's bus gear, SW_ARM bracket and cup gear. Lower the posts with the idler onto the deck about 4 mm to the back (-x) of their place, then slide them forward so the small pinion meshes the drum A segment teeth, and bolt them down (2 x M3 x 12 each; the front screw of post 2 needs the short arm of the hex key). Turn drum A back to LAUNCH (mark L at the top): the flat-sided lock cam must sit in the drum lock ring and the idler can no longer turn by hand.

Parts: drum_idler_gear, drum_idler, drum_bracket_idler, drum_bracket_idler2.

Step 13 (main_shaft 145). Drum: shift rail

Slide the shift rail in from the outboard end, over the idler posts, and press it into bracket A (flat facing forward).

Parts: drum_rail.

Step 14 (main_shaft 146). Main shaft: coupler on its hub (show build: wiper not linked)

Push the coupler hub onto the wiper output shaft (flats in line) until it stops on the end of the flats; its lower end goes into the puck bore, just clear of the motor's output boss. Drop the M8 nut into the hub pocket onto the thread (it sits about 8 mm down the pocket) and tighten it with a 13 mm socket (1/4 inch drive). Slide the outer coupler over the hub. Show build: NO shear pin, so the coupler (and the main shaft) turn free on the hub: the hub on the unwired motor shaft is only the bearing of the main shaft's motor end. Then slide the REEL gear onto the coupler journal (plain face first) and screw the retaining washer to the journal end (2 x M3 x 6): the gear must spin free.

Parts: drum_coupler_hub, drum_coupler, drum_gear_reel, drum_reel_washer.

Step 15 (reel_capstan 150). Screw the reel wall to the deck

Stand the reel wall on the deck at its marked place (its long foot reaches back onto the deck extension tile). Put nine M3 screws down through the counterbored foot holes into the deck inserts and tighten. Do this first: the gears later cover some of the holes.

Parts: reel_wall.

Step 16 (reel_capstan 151). Sleeve, bus gear, swing arm and swing idler

Push the sleeve into the big hole in the wall from the front, its rib in the slot, flange flat on the wall, and screw it on with three M3 x 8 screws (they tap into the wall). From behind slide the 46-tooth bus gear onto the sleeve, gear face first, hub toward you, and snap the large C-clip into the groove behind the hub. Slip the swing arm's ring over the bus gear's hub, idler pin pointing at the wall, and put the 30-tooth swing idler on the pin. The arm swings 12 degrees between its two stop pins.

Parts: reel_sleeve, reel_sleeve_clip, reel_bus.

Step 17 (reel_capstan 152). Cup shaft, cup gear and riser gear

Push the cup shaft through the sleeve from BEHIND, flat end first, until it comes out in front of the wall. Behind: slide the 24-tooth cup gear onto the flat (hub forward, raised track toward you) and snap its C-clip on. In front: slide the 30-tooth riser gear onto the flat and snap its C-clip on. The cup shaft must turn freely.

Parts: reel_k, reel_gk, reel_k_clip, reel_gk1, reel_k_clip2.

Step 18 (reel_capstan 153). Riser idlers and K2

Push the two idler axle pins through the 34-tooth and the 30-tooth riser idlers and press them into their holes in the wall front (34 teeth next to the cup shaft gear). Push the K2 stub through its wall bushing from the front with its 30-tooth gear and the 18-tooth bevel on the flats, the bevel teeth facing you, and snap its C-clip on behind the wall. Turn the cup gear: the whole riser turns.

Parts: reel_ik1, reel_ik1_axle, reel_ik2, reel_ik2_axle, reel_k2, reel_gk2, reel_bk2.

Step 19 (reel_capstan 154). Countershaft C and the REEL idler

Slide the 20-tooth C bevel onto the front of countershaft C (teeth facing away from the shaft), feed the shaft through its wall bushing from the front, and from behind slide the 22-tooth C gear onto the flat and snap its C-clip on. Push the REEL idler axle through the 41-tooth idler and press it into the wall from behind: the idler meshes the drum's REEL gear and the C gear.

Parts: reel_cs, reel_bc, reel_gc, reel_cs_clip, reel_ic, reel_ic_axle.

Step 20 (reel_capstan 155). Clutch hub

Press one 608 bearing into each end of the clutch hub. Lay the friction disc on the front, its slots on the three lugs, raised ring outward. Slide the hub onto the reel shaft from the shaft's thin end until the disc meets the shaft's round head. Slide the thrust ring onto the shaft behind the hub, small boss toward the bearing.

Parts: reel_hub, reel_hub_flange, reel_brg_1, reel_brg_2, reel_s, reel_t.

Step 21 (reel_capstan 156). Brake crank, rocker and bar

Grease the crank boss low on the left of the wall front. Slide the black bell crank onto it: the short arm lies in the recess in the wall and points right, the long arm sticks straight out from the wall. Put the printed washer on the boss and screw the M3 x 20 into it: the crank must swing freely. Screw the M3 x 10 FIRM screw into the rocker's leg from the front with its nut on the front face, until 0.35 mm of its tip shows at the back. Hang the yellow rocker between the two ears at the top with an M3 x 25 screw and nut (snug, it must swing); its round hole goes over the small boss, its leg rests on the small pad on the wall, the FIRM screw tip just above the crank's short arm. Screw two M3 x 8 screws into the rocker's two bosses from behind, until their tips stand 0.5 mm proud of the front. Lay the brake bar in the two U guides, its raised lug against the crank's long arm, nose toward the drum. Close each guide with its cap and two M3 x 16 screws. Push the nose: the crank swings and the rocker leg moves away from the wall; let go and it all springs back.

Parts: reel_crank, reel_crank_washer, reel_crank_screw, reel_firm_screw, reel_rocker, reel_rocker_pin, reel_trim_screws, reel_bar, reel_bar_cap_1, reel_bar_cap_2, reel_bar_screws.

Step 22 (reel_capstan 157). Reel shaft into the wall

Feed the reel shaft's thin end through the rocker's hole and the wall bushing from the front. From behind, slide the 20-tooth reel gear onto the D-flat, hub toward the wall, and snap its C-clip on. Swing the arm toward the reel gear (to its stop): the idler meshes it.

Parts: reel_gr, reel_s_clip.

Step 23 (reel_capstan 158). T2 take-off

Push the T2 stub through its wall bushing from the front with the 40-tooth T2 gear and the 20-tooth T bevel on its flats (bevel teeth facing you): the T2 gear meshes the clutch hub's teeth. Snap its C-clip on behind the wall. Turn the clutch hub by hand: T2 turns the same speed.

Parts: reel_t2, reel_gt2, reel_bt2.

Step 24 (drum_shift 160). Drum: fork and dog contacts

Battery unplugged. Press an M3 nut into the hex hole of the yoke tab. Thread an M3 x 8 into each dog post boss first, heads toward each other (M3 x 8: a longer screw reaches the REEL gear; no room for a driver later). Turn the drum to about 180. Slide the fork sleeve onto the rail, yoke facing the main shaft, and swing it about 20 deg pin end away from the drum while the dog post slides on (its bar reaching back over the yoke tab; fit it before cad-reel's pulley 2). Grease the rail. Tighten the dog post clamp screw (M3 x 8, tapped) onto the rail flat. Battery unplugged: nut wire to GND, REEL side screw to SW_DOG_R, LAUNCH side screw to SW_DOG_L.

Parts: drum_fork.

Step 25 (drum_shift 161). Drum: drum B, hand lever, bracket B

Swing the fork up on the rail, slide drum B onto the axle (hex socket over the drum A spigot; it fits one way only, cut corner to filled corner), slide the fork to its middle position and swing it down so its pin drops into the groove (drum near 180). Slide the hand lever onto the hex boss. Press bracket B onto the axle and rail ends and bolt it down (2 x M3 x 12). The drum turns by hand with about 0.6 mm end play.

Parts: drum_drum_b, drum_hand_lever, drum_bracket_b.

Step 26 (drum_shift 162). Drum: N20

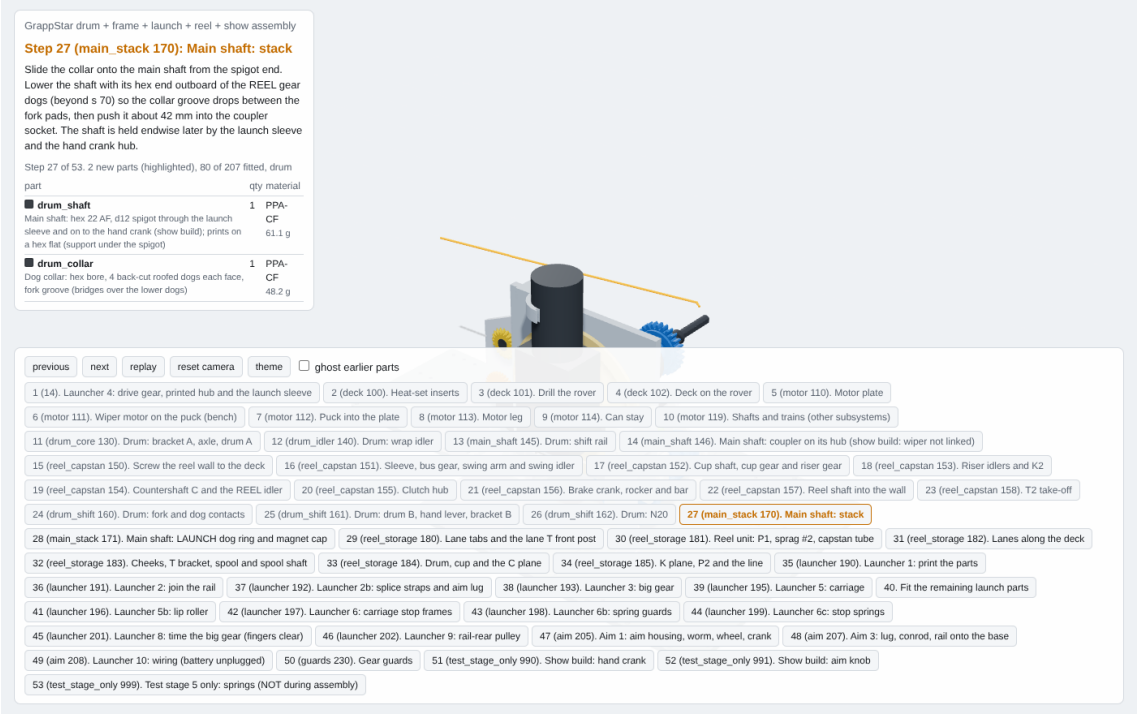
Battery unplugged. Show build: no pinion, the N20 is not linked (the drum turns by its hand lever). Slide the N20 into the cradle from the open end, fit the end plate (2 x M3 x 8). Press an M3 nut into the pocket in the bracket A ear front face, wire along its groove. Screw the cradle to the ear from the cradle side: pivot screw M3 x 12 through the round hole, swing screw M3 x 12 through the arc slot at its full-mesh end (the slot only lets the cradle swing out), both tapped into the ear. Turn the SW_N20 screw (M3 x 8, tapped in the cradle plate) in until its tip just touches the nut. Battery unplugged: nut to GND, screw to SW_N20.

Parts: drum_n20, drum_n20_cradle, drum_n20_strap.

Step 27 (main_stack 170). Main shaft: stack

Slide the collar onto the main shaft from the spigot end. Lower the shaft with its hex end outboard of the REEL gear dogs (beyond s 70) so the collar groove drops between the fork pads, then push it about 42 mm into the coupler socket. The shaft is held endwise later by the launch sleeve and the hand crank hub.

Parts: drum_shaft, drum_collar.



Step 28 (main_stack 171). Main shaft: LAUNCH dog ring and magnet cap

Fit the LAUNCH dog ring on the launch sleeve flange (sleeve lugs in the ring slots, 3 x M3 x 10 from the sleeve side into the ring) before the launch sleeve goes on the spigot (launcher steps). Show build: no magnet cap; the hand crank on the spigot end holds the shaft endwise (show steps).

Parts: drum_dogring_launch.

Step 29 (reel_storage 180). Lane tabs and the lane T front post

Screw the lane C tab and the lane K tab to the wall front, each with two M3 x 10 (they tap into the wall). Screw the lane T front post to the deck extension with two M3 screws into the inserts.

Parts: reel_tab_c, reel_tab_k, reel_post_t.

Step 30 (reel_storage 181). Reel unit: P1, sprag #2, capstan tube

Screw the P1 leg to the deck extension and bolt P1 onto it (one M3 x 16). Slide the capstan tube in from the FRONT, D-flat end first, until its collar sits on P1's bushing.

Parts: reel_p1, reel_p1_leg, reel_c.

Step 31 (reel_storage 182). Lanes along the deck

Slide each bevel onto its lane's front end (teeth toward the front). Feed lane C (d12) from the front through the lane C tab and P1, its bevel meshing the C bevel; lane K (d8) through the lane K tab and P1, its bevel meshing the K2 bevel; lane T (d8) through the front post and the P1 leg, its bevel meshing the T2 bevel. Slide the lane T rear bevel onto lane T (teeth toward the front).

Parts: reel_lane_c, reel_bcx, reel_lane_k, reel_bkx, reel_lane_t, reel_btx, reel_bt3x.

Step 32 (reel_storage 183). Cheeks, T bracket, spool and spool shaft

Press a 608 into each cheek. Screw both cheeks and the T bracket to the deck extension (M3 into the inserts). Hold the spool between the cheeks and push the spool shaft from the inboard side through the T bracket, the inboard 608, the spool (flat in the flat) and the outboard 608; slide the 16-tooth gear onto its inboard end. Put T3 with its bevel and 16-tooth gear into the T bracket: its bevel meshes the lane T rear bevel. Press the idler axle through the 34-tooth take-up idler into the bracket so it meshes both 16-tooth gears.

Parts: reel_cheek_i, reel_cheek_o, reel_brg_3, reel_brg_4, reel_t_bracket, reel_spool, reel_ss, reel_gs, reel_t3, reel_bt3, reel_gt3, reel_ti, reel_ti_axle.

Step 33 (reel_storage 184). Drum, cup and the C plane

Slide the green drum onto the front of the capstan tube, flange first, key slot on the rib. Push the rear cup shaft through the tube from the FRONT, plain end first. Slide the wrap cup onto its front end, flat on the flat, eyelet at the BOTTOM, and push Harry's small C-clip onto the neck. Behind P1: slide G_c (20 teeth, m2) onto the tube's flat and clip it; slide G_a onto lane C's flat. Press sprag #2 (CSK20PP) into the 58-tooth idler's hub, key slot on the printed rib. Check the arrow: the idler must turn freely the way it turns when the capstan winds in and lock the other way (turn the capstan by hand once assembled: wind-in free, payout locked). Push the keyed I_c axle from the front through P1 and the sprag (key in the keyway, flat in P1's flat) and clip it behind the idler.

Parts: reel_drum, reel_kc, reel_cup, reel_cup_clip, reel_gc2, reel_c_clip, reel_ga, reel_icr, reel_icr_axle.

Step 34 (reel_storage 185). K plane, P2 and the line

Battery unplugged. Slide G_k (20 teeth) onto the rear cup shaft and G_a' onto lane K. Press the I_k axle through the 76-tooth idler into P2's front face. Put the three spacer tubes on P1's screws, offer P2 up so the cup shaft and lane K enter its journals and I_k meshes both gears, and bolt it to the spacers, its leg and the two cheeks; clip the cup shaft behind P2. Turn the cup to home (eyelet at the bottom). Wind the line onto the spool (line over the TOP, leaving toward the front), lead it under the capstan and through the cup eyelet from behind, then straight forward to the pulley at the rail's rear end. Set LIGHT with the drum at LAUNCH: turn both trim screws the same amount until 25 g hung on the line end just pulls line off slowly (0.25 N) and 20 g does not. Set FIRM last, with the drum in REEL: the line must slip at 5.0 N on the luggage scale (FIRM screw in = more).

Parts: reel_gkc, reel_ga2, reel_ikr, reel_ikr_axle, reel_spacer_1, reel_spacer_2, reel_spacer_3, reel_p2, reel_p2_leg, reel_kc_clip.

Step 35 (launcher 190). Launcher 1: print the parts

All PLA unless named. PPA-CF: big gear, drive gear, launch sleeve, big gear pin, SW_SEATED plunger and pushrod, and the aim gears (sun, idler, output, bevel, worm, crank). Use each part's print orientation from the print sheet: the carriage body lies on its side across the bed diagonal; the stop springs print upside down at 100 percent infill (all perimeters); rail pieces lie on their largest flat face. 0.2 mm layers, 4 walls on every loaded part. Hardware (launcher and aim train): M3 screws 2 x 6, 3 x 8, 14 x 10, 23 x 12, 5 x 16, 15 x 20, 4 x 30, plus 4 x 10 for the aim housing to the deck (Aim 3); 10 M3 nuts, 4 M3 nylocs; M8 x 25 bolt and M8 nut; 2 x 608; 4 x 693ZZ (3 tether pulleys, 1 lip roller); 2 x 8 mm steel rod pins (cut to the length on the print sheet) for the spring anchors; sprag #1 (CSK20PP); the two springs.

Step 36 (launcher 191). Launcher 2: join the rail

Lay the rear, mid and front rail pieces in a line on a flat table. Push each joint together and fit 2 x M3 x 20 with nuts through each flange pair (8 in total). Tighten evenly; file any step inside the track where the pieces meet so the carriage shoes slide over it.

Parts: launch_rail_rear, launch_rail_mid, launch_rail_front.

Step 37 (launcher 192). Launcher 2b: splice straps and aim lug

Screw a splice strap on each wall top across the front joint (4 x M3 x 10 self tap each). Bolt the aim lug to the left rail wall now (2 x M3 x 12 self tap): the spring guards cover its screws later.

Parts: launch_rail_splice_l, launch_rail_splice_r, launch_aim_lug.

Step 38 (launcher 193). Launcher 3: big gear

Hold the big gear between the two cheeks of the mid piece, the toothed sector towards the drum side (right) and the full ring of teeth towards the left, with the spacer washer between the gear and the left cheek. Push the big gear pin through cheek, gear, washer, cheek from the right and snap the C clip into its groove. The gear must spin freely with almost no end play.

Parts: launch_big_gear, launch_gear_pin, launch_gear_clip, launch_gear_washer.

Step 39 (launcher 195). Launcher 5: carriage

Turn the big gear so its toothless gap faces the rack. Slide the carriage BODY into the track from the FRONT end (the tether fork blocks the rear), rack teeth down, until its rear face reaches the stop position. Yoke: push the SW_SEATED pushrod into the block bore from the rear, collar first, then push the plunger's pin up through the block from below (pad under the left arm) and press the head cap on the pin top. Lower the yoke onto the carriage rear so its key plates drop into the slots in the body webs, and fit 1 x M3 x 10 per side through the yoke into the webs. Press a printed spring peg up into each inner (y 38) peg hole from below, flat to the rear, to the end of the hole.

Parts: launch_carriage, launch_carriage_yoke, launch_yoke_peg_l, launch_yoke_peg_r.

Step 40. Fit the remaining launch parts

Parts without an explicit assembly step: launch_dart.

Parts: launch_dart.

Step 41 (launcher 196). Launcher 5b: lip roller

Press a 693ZZ into the lip roller, put a printed thrust washer each side in its bore (hub to the bearing), drop roller and washers into the fork at the carriage front and fit the M3 x 16 and a nyloc nut; it must spin freely. The tether runs over it in tow and retrieval, and the dart's tail slides in over it through the funnel onto the cradle.

Parts: launch_muzzle_roller, launch_muzzle_roller_brg, launch_muzzle_roller_axle, launch_muzzle_washer_l, launch_muzzle_washer_r.

Step 42 (launcher 197). Launcher 6: carriage stop frames

For each side, before any spring: slide the stop frame onto the rail from the outboard side (it does not drop straight down past the inboard preload ledge). 2 x M3 x 12 through the rear foot into the rail wall, and 2 x M3 x 16 down through the front foot into the rail front plate, driven through the access holes in the frame's front wall.

Parts: launch_stop_frame_l, launch_stop_frame_r.

Step 43 (launcher 198). Launcher 6b: spring guards

Screw the spring guard plates to the rail walls and anchor posts (M3 self tap into the pilots, see the print sheet), four per side: a (rear, with the end wall), b1 and b2 (either side of the stop frame's rear leg) and c (front, 2 x M3 x 12 into the front plate edge). Slide each one in from the side, under the stop frame's outboard bar. Wall brackets: M3 x 12 self tap, head and hex key through the hole in the plate; b2 foot: M3 x 10 down. Never cock with a guard off.

Parts: launch_guard_l_a, launch_guard_l_b1, launch_guard_l_b2, launch_guard_l_c, launch_guard_r_a, launch_guard_r_b1, launch_guard_r_b2, launch_guard_r_c.

Step 44 (launcher 199). Launcher 6c: stop springs

Put each stop spring on its ledges inside its frame, rear bar towards the yoke arm, and pull it onto the frame's front wall with the 2 x M3 x 20 draw screws (heads on the front wall), a turn each in turn, until it sits on the wall (6.6 mm of preload). The rear bar must rest on both ledges.

Parts: launch_stop_spring_l, launch_stop_spring_r.

Step 45 (launcher 201). Launcher 8: time the big gear (fingers clear)

No springs fitted yet. Hold the carriage on its stop by hand and turn the drive gear FORWARD until the first tooth after the toothless gap just touches the rearmost rack tooth: this is HOME (SW_GAP is set to it in the Sensors steps). Turning on pulls the carriage back smoothly; at full cock the cut last tooth lets go (push the carriage home by hand). Hold the drive gear from the side and never put a hand in front of the carriage.

Step 46 (launcher 202). Launcher 9: rail-rear pulley

Press a 693ZZ into the printed pulley. Fit it in the fork at the rail's rear end with a printed top-hat washer on each side (hub toward the bearing) and push the M3 x 20 axle through (nyloc nut on the far side, snug: the pulley must spin). D50: this is the only pulley on the tether. The line comes from the capstan eyelet behind the launcher, runs UNDER this pulley and forward along the rail axis through the pusher hole to the dart's tail cap. Never fire without the tether attached (D11).

Parts: launch_pulley_top, launch_pulley_top_brg, launch_pulley_top_axle, launch_pulley_top_washer_l, launch_pulley_top_washer_r, launch_line_p, launch_line_a0, launch_line_a1, launch_line_a2, launch_line_a3, launch_line_a4, launch_line_a5, launch_line_a6, launch_line_a7, launch_line_a8.

Step 47 (aim 205). Aim 1: aim housing, worm, wheel, crank

Build the aim housing on the bench (it goes onto the base later, after the rail). The AS5600 carrier and SW_RAIL_LOW are cad-sensors parts (Sensors steps), screwed into the pilots on the housing. Push the countershaft in from the inboard side through the inner wall bushing, through the big bevel (D bore, slid on from the outboard end of the shaft) and into the outer wall bushing. Drop the M8 nut into the hex pocket on the inner face of the outer wall, fit the two 608 bearings into the crank (worm wheel) hub, hold the crank in place and push the M8 x 25 bolt in from outboard through the wall, the nut and both bearings; tighten so the head and nut clamp the wall. Lay the worm shaft with the small bevel in its saddles and screw the cap on (4 x M3 x 30 self tap). The housing screws to the deck later (Aim 3), from the top: 4 x M3 x 10 down through the counterbored holes in its foot flanges (two behind the spine, one in front of the outer wall, one on the inboard tab) into the deck's heat-set inserts. The SW_RAIL_LOW bracket on the foot extension has its own deck screw (Sensors 2).

Parts: launch_aim_housing, launch_aim_cap, launch_aim_worm, launch_aim_crank, launch_aim_bolt, launch_aim_brg_1, launch_aim_brg_2, launch_aim_output, launch_aim_bevel30, launch_aim_nut.

Step 48 (aim 207). Aim 3: lug, conrod, rail onto the base

With the sun and swing arm already on the sleeve (the lug is on the rail since Launcher 2b), slide the rail assembly's sleeve along the main axis onto the drum's d12 spigot from the outboard side (the LAUNCH dog ring is already bolted to its flange, Launcher 4) until the ring's dogs sit by the drum collar. Hang the conrod's upper eye on the lug pin and snap a printed C clip into that groove. Then slide the aim housing on from the outboard side so the sleeve end enters its bushing and the crank pin enters the conrod's lower eye (hold the swing arm with the idler pointing down and a little to the rear, clear of the output gear, and let the arm go in between the two stop posts on the housing's inner face as they come up to it); snap a printed C clip onto the crank pin and screw the housing to the deck from the top: 4 x M3 x 10 through the counterbored foot flange holes into the deck inserts. Turn the sleeve backward by hand: the arm swings the idler into the output gear, comes to rest on its engaged stop post and drives the gear without binding; forward, the arm swings out onto the other post and the output gear stays still. Then turn the wiper backward slowly: the rail rises to 47 deg, then lowers to 6 deg.

Parts: launch_aim_conrod, launch_aim_crankpin_clip, launch_aim_lugpin_clip.

Step 49 (aim 208). Launcher 10: wiring (battery unplugged)

Battery unplugged. Run the SW_HOME, SW_SEATED and SW_GAP wires along the rail to the pivot and leave a 150 mm loop at the main axis before they go to the frame. No wire goes onto the carriage. Each contact: one wire to its ESP32 pin (SW_HOME 21, SW_GAP 39, SW_RAIL_LOW 40, SW_SEATED 41, as params.SWITCHES), the other to GND.

Step 50 (guards 230). Gear guards

After every train has been turned by hand: the hood goes over the gears in front of the motor plate, M3 x 10 through each lug into the plate inserts; the front guard stands on the deck between the drum gears and the bay, M3 x 10 into the deck inserts. Turn the wiper by hand once: nothing rubs.

Parts: frame_guard_hood, frame_guard_front.

Step 51 (test_stage_only 990). Show build: hand crank

After the launcher: slide the hand crank onto the main shaft spigot end, hub first, flat to flat, until the hub is about 2 mm from the launch sleeve end, and tighten the M3 x 6 onto the flat. Press the handle peg into the arm hole (a drop of glue if loose). Turning the crank turns the main shaft and whatever the drum has selected: LAUNCH cocks the launcher, REEL winds the capstan.

Parts: show_hand_crank, show_crank_handle.

Step 52 (test_stage_only 991). Show build: aim knob

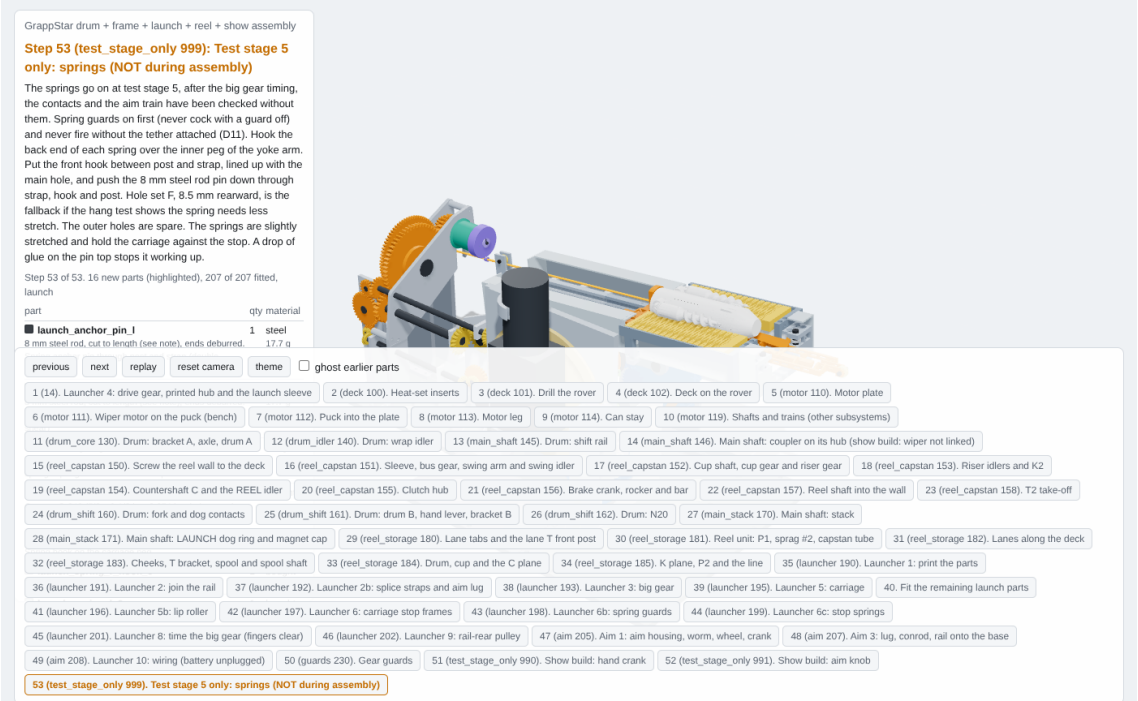
Push the aim knob onto the aim countershaft end outside the housing, flat to flat (a drop of glue). Turning it raises or lowers the rail through the bevels, the worm and the crank-rocker; the worm holds the angle.

Parts: show_aim_knob.

Step 53 (test_stage_only 999). Test stage 5 only: springs (NOT during assembly)

The springs go on at test stage 5, after the big gear timing, the contacts and the aim train have been checked without them. Spring guards on first (never cock with a guard off) and never fire without the tether attached (D11). Hook the back end of each spring over the inner peg of the yoke arm. Put the front hook between post and strap, lined up with the main hole, and push the 8 mm steel rod pin down through strap, hook and post. Hole set F, 8.5 mm rearward, is the fallback if the hang test shows the spring needs less stretch. The outer holes are spare. The springs are slightly stretched and hold the carriage against the stop. A drop of glue on the pin top stops it working up.

Parts: launch_anchor_pin_l, launch_anchor_pin_r, launch_spring_l_hook_a, launch_spring_r_hook_a, launch_spring_l_hook_c, launch_spring_r_hook_c, launch_spring_l_0, launch_spring_l_1, launch_spring_l_2, launch_spring_l_3, launch_spring_l_4, launch_spring_r_0, launch_spring_r_1, launch_spring_r_2, launch_spring_r_3, launch_spring_r_4.



6. Audit summary

Result: CLEAN

207 parts checked over 46300 states.

Allowed contacts: 92.